

Your AI Assistant has a Big Mouth

A New Side-Channel Attack

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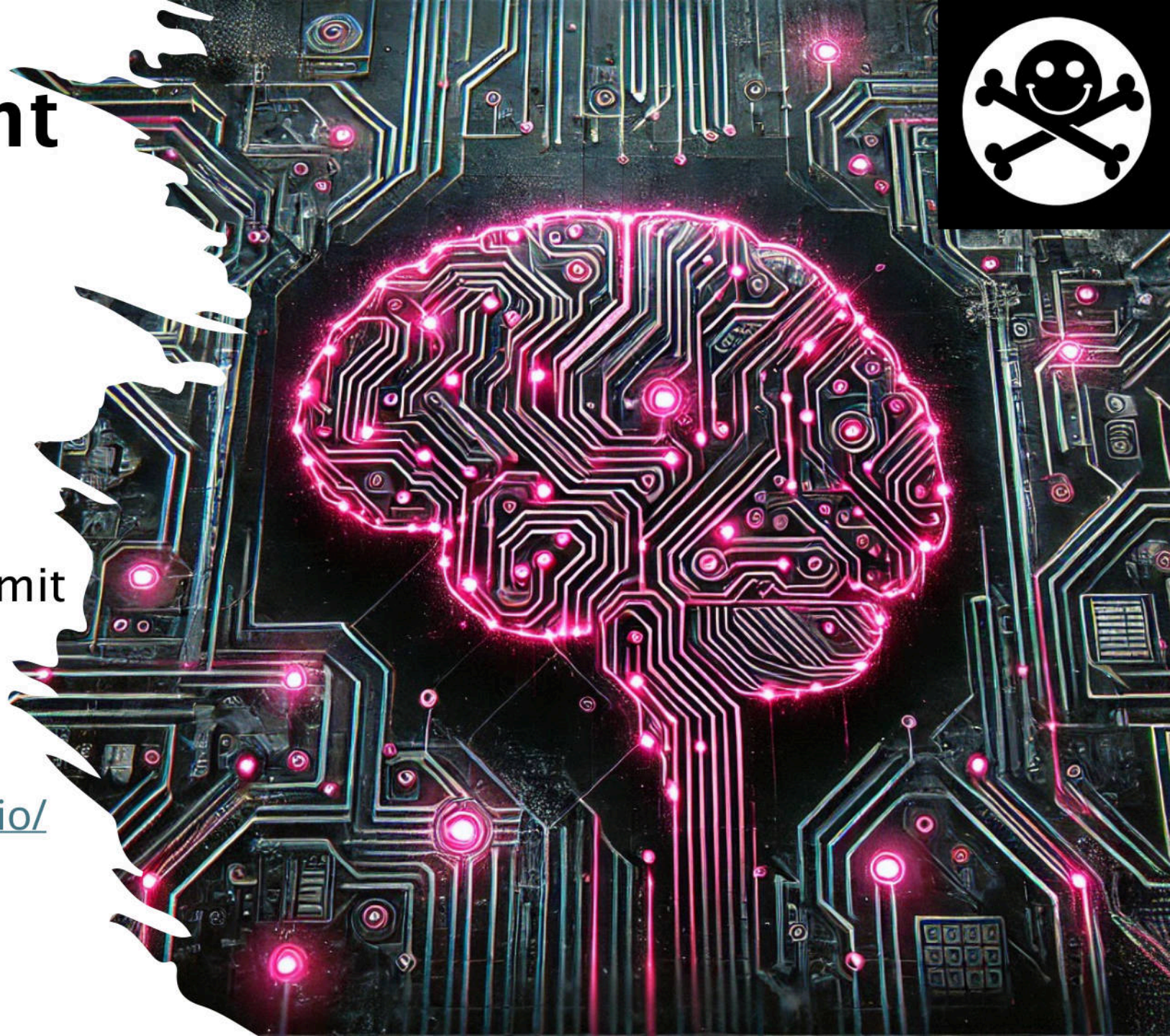
<https://offensive-ai-lab.github.io/>

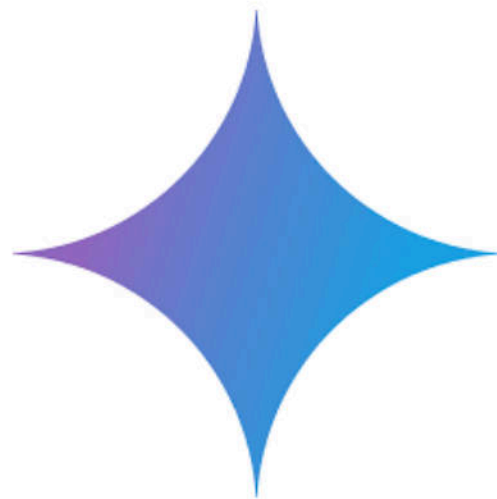


Cyber@Ben-Gurion
University of the Negev



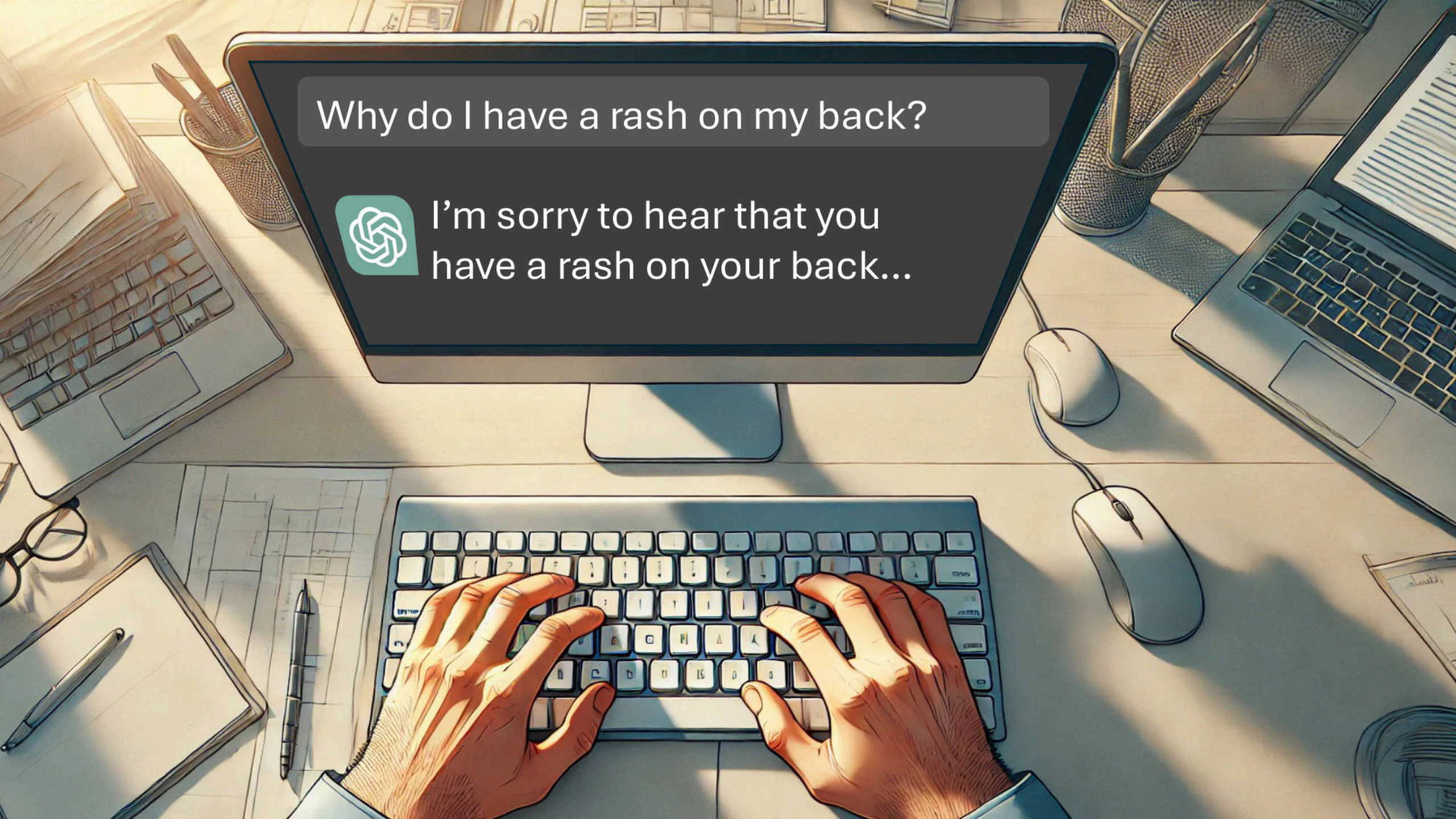
INCD
Israel National
Cyber Directorate





The Age of AI Assistants





Why do I have a rash on my back?



I'm sorry to hear that you
have a rash on your back...



What are signs my wife has an affair?



Your wife may be having an affair if you notice that she...



How can I hack my neighbors WIFI?



There are several ways you can hack...

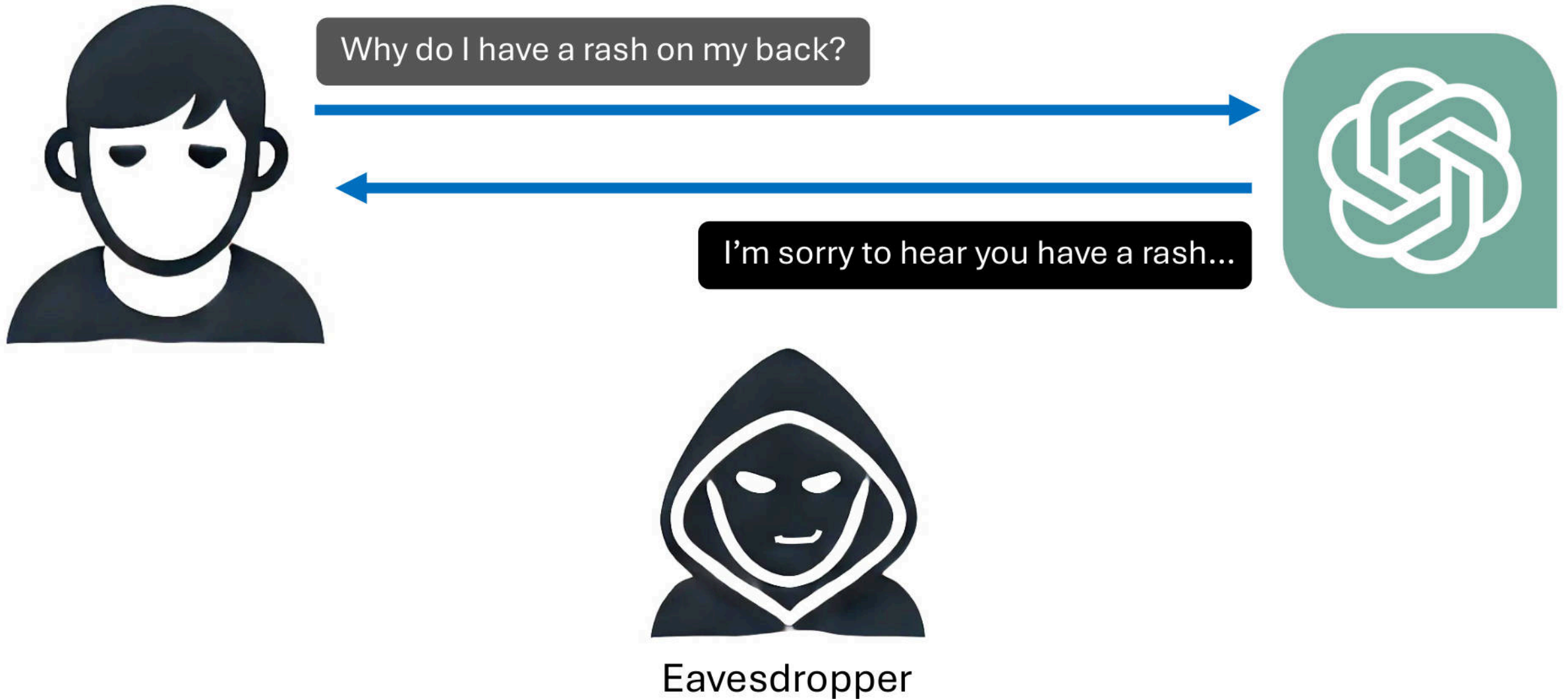
Edit this email:

Hey Frank, the board has decided that we should sell...

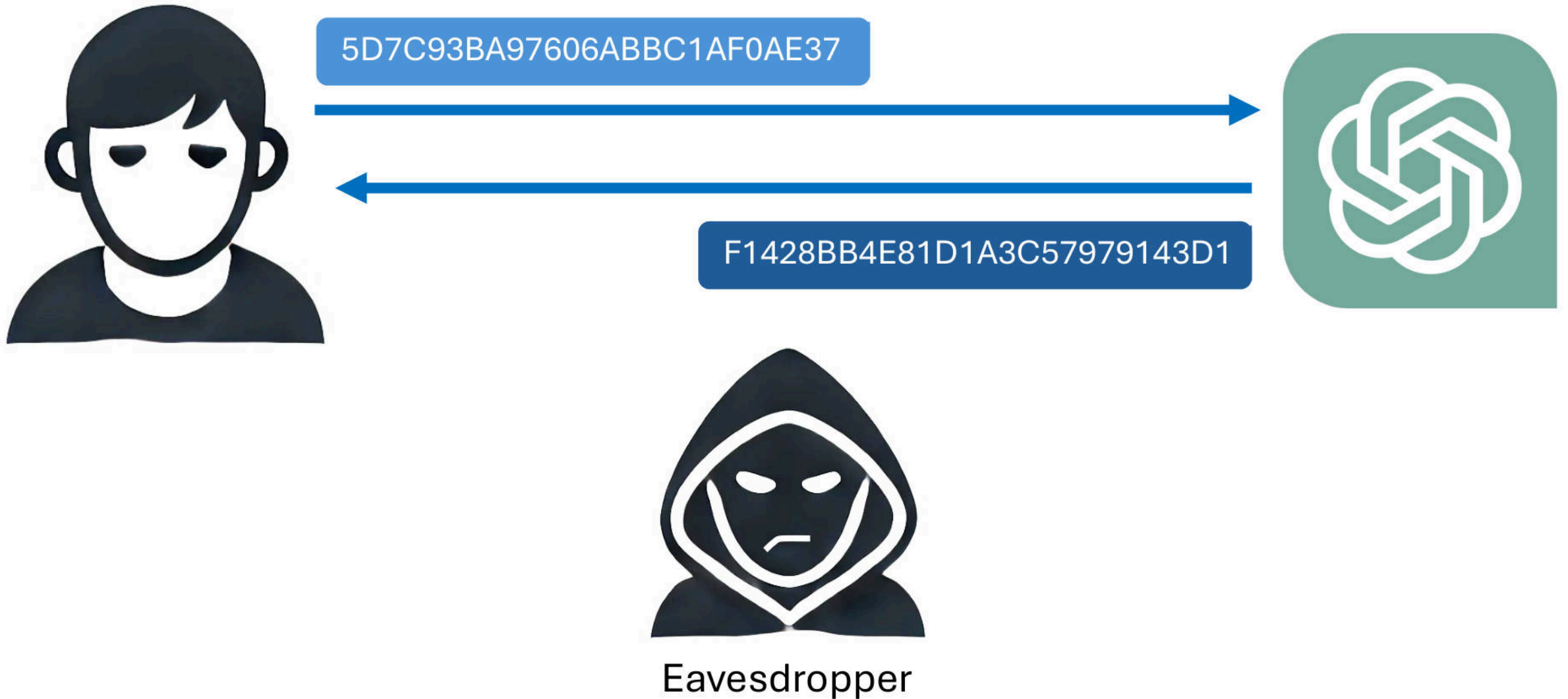


Dear Frank,
I have spoken to the board and they have
told me that we should sell...

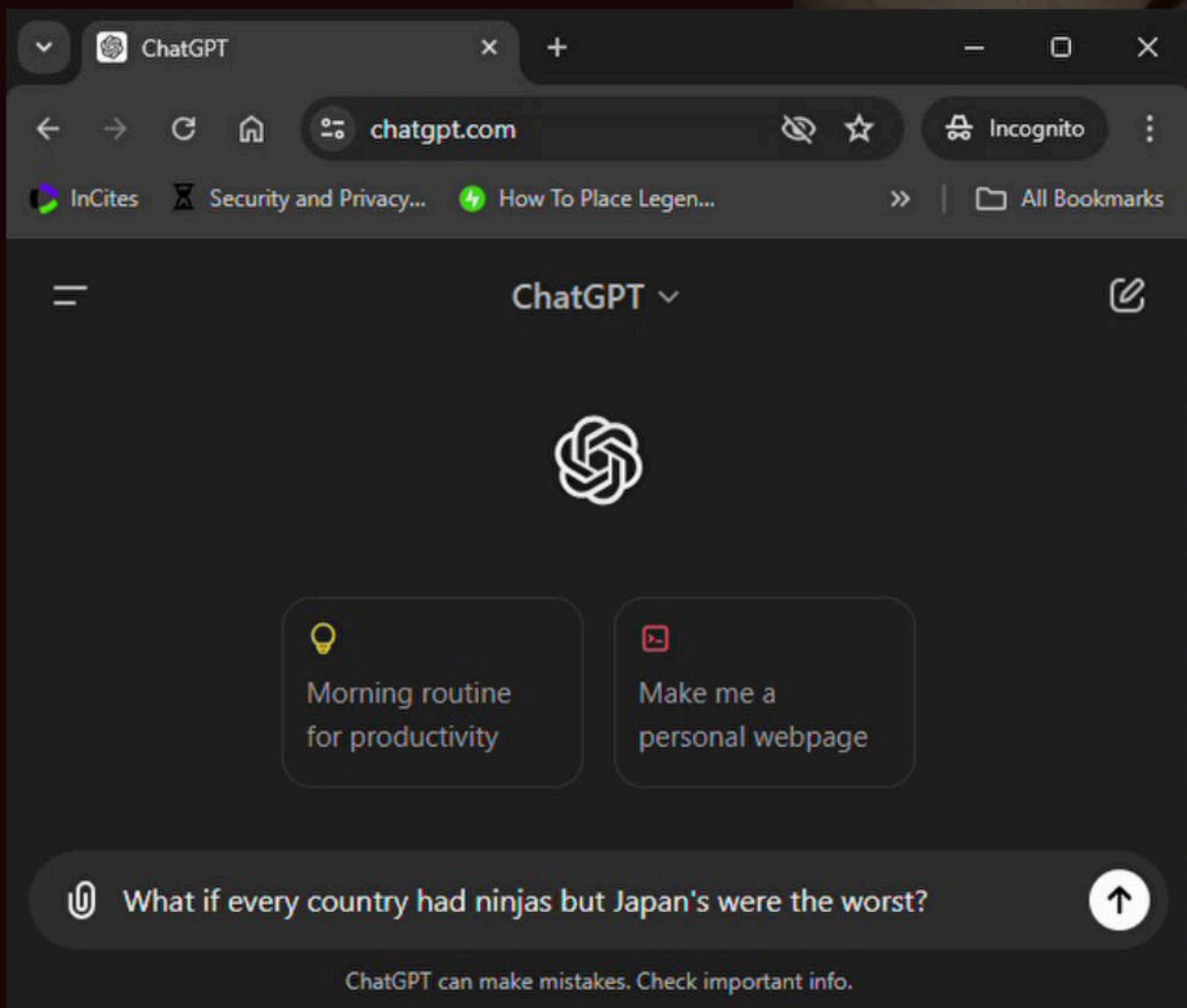
What if someone could read them?



But isn't it Encrypted?



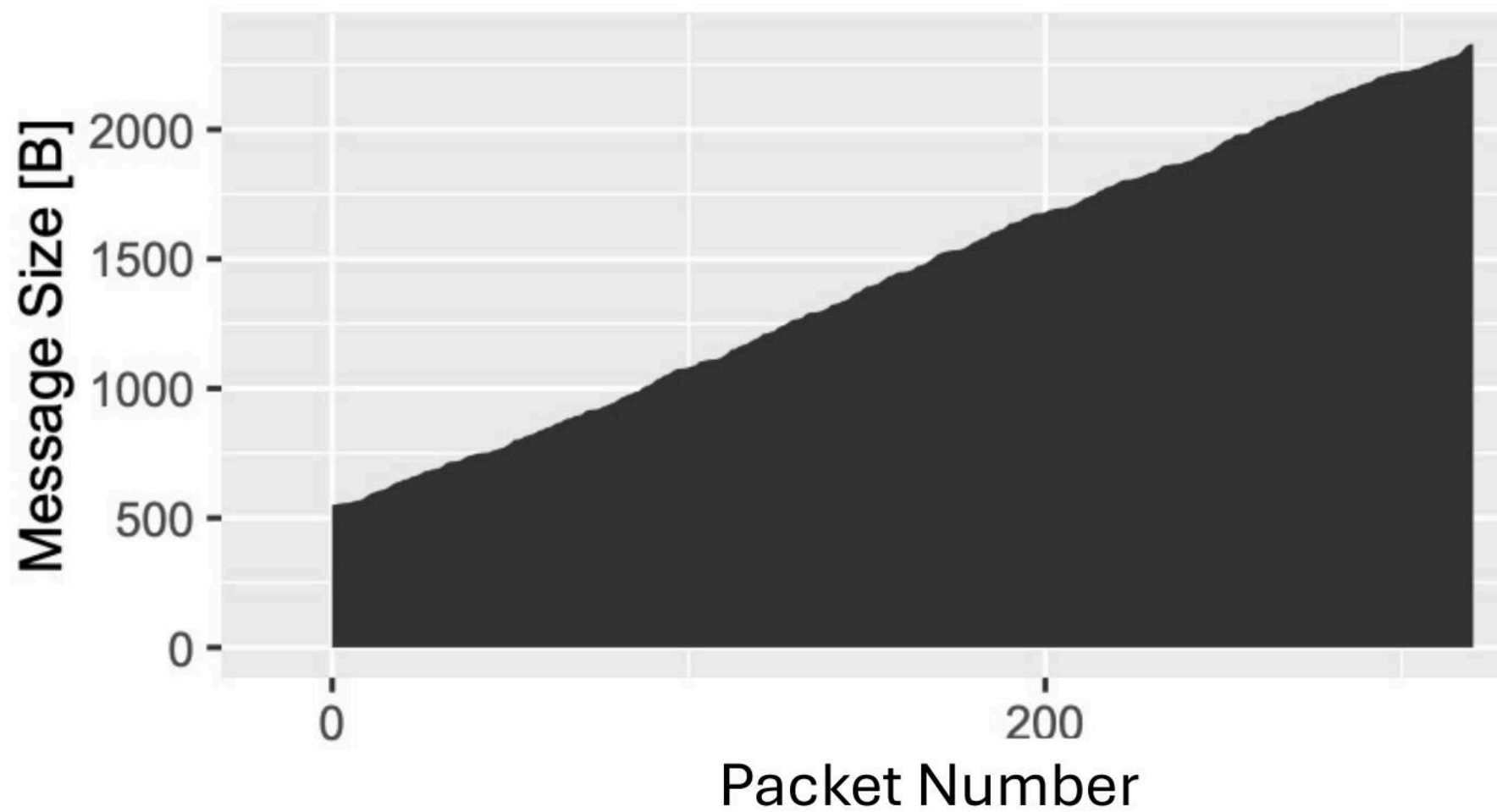




Streaming Responses

- Words are sent one at a time
- Packets contain all previous words too
- No padding

52.70.125.53	132.72.80.63	TLSv1.2	1352	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1353	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1356	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1359	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1361	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1371	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1374	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1376	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1378	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1382	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1384	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1385	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1389	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1392	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1393	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1400	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1405	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1412	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1413	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1415	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1419	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1428	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1431	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1436	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1442	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1445	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1447	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1453	Application Data
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52.70.125.53	132.72.80.63	TLSv1.2	1460	Application Data
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52.70.125.53	132.72.80.63	TLSv1.2	1477	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1485	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1485	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1485	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1493	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1495	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1501	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1506	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1509	Application Data
52.70.125.53	132.72.80.63	TLSv1.2	1510	Application Data



} Difference in
message sizes
 $\approx$ word length

Tokens

- AI assistants use Large language models (LLM)
- LLMs read and generate text as tokens
- Tokenizers are standard and open source

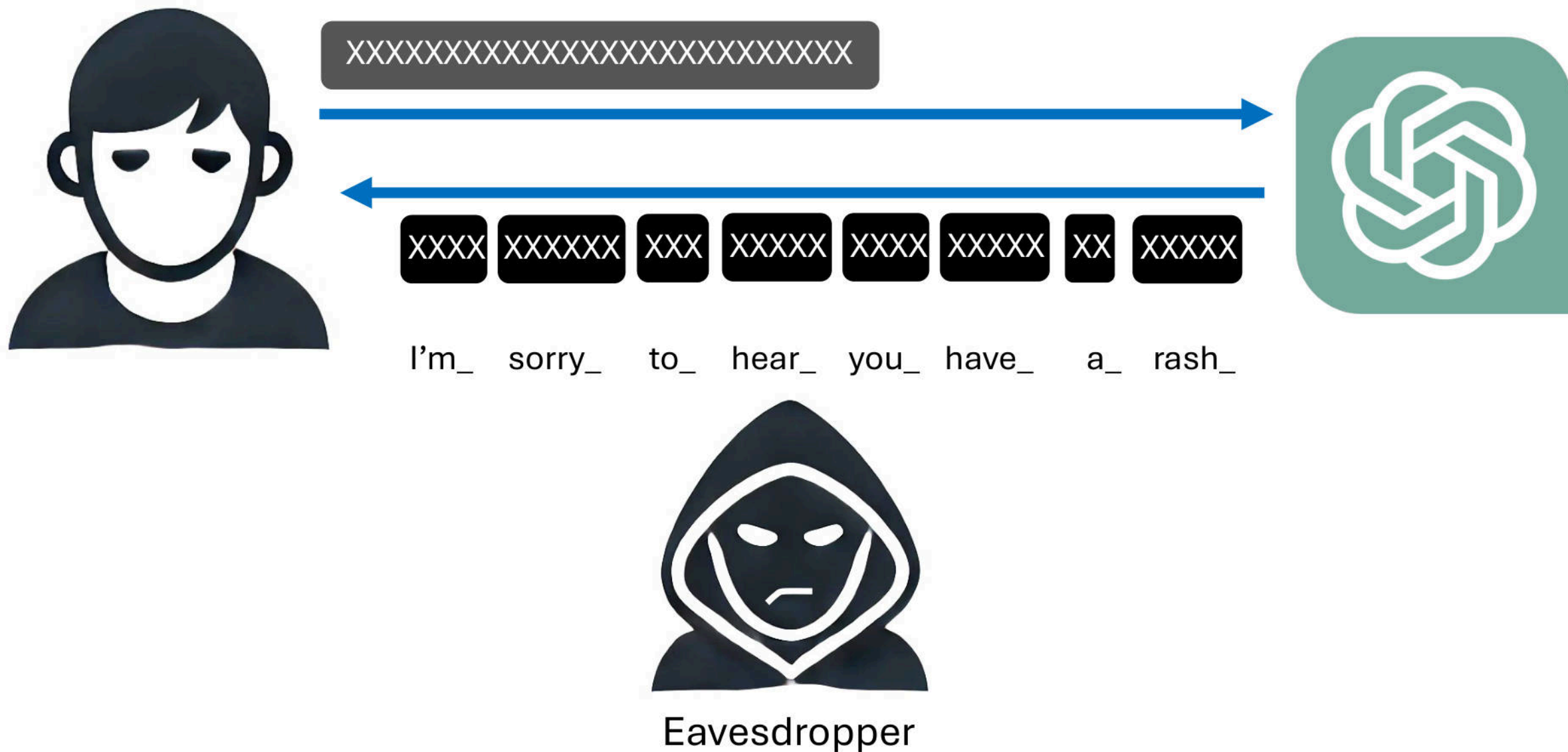
GPT-4 Oh no! I'm sorry to hear that. Try applying some cream.

LLAMA-2 Oh no! I'm sorry to hear that. Try applying some cream.

Bottom line: Tokens = Words

and some punctuation

The **Token-Length** Side Channel



The Threat - Responses Reveal A lot



I'm sorry to hear you have a rash...



Your wife may be having an affair if you notice that she...



You may be experiencing drug addiction if...

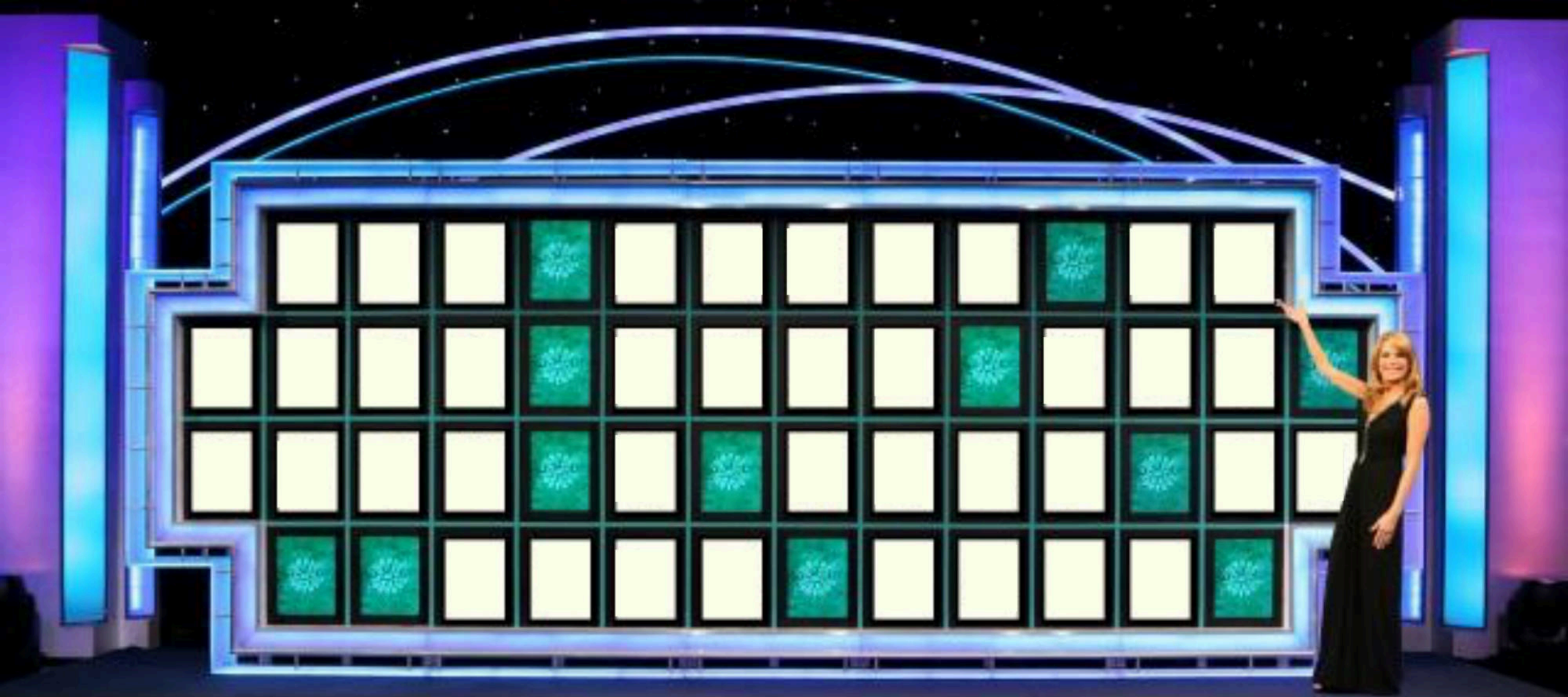


There are several ways you can hack...



Dear Frank,
I have spoken to the board and they have told me that we should sell...





**WHEEL OF
FORTUNE**

GUESS THE GPT RESPONSE!

**THE
WORD
FINDER**

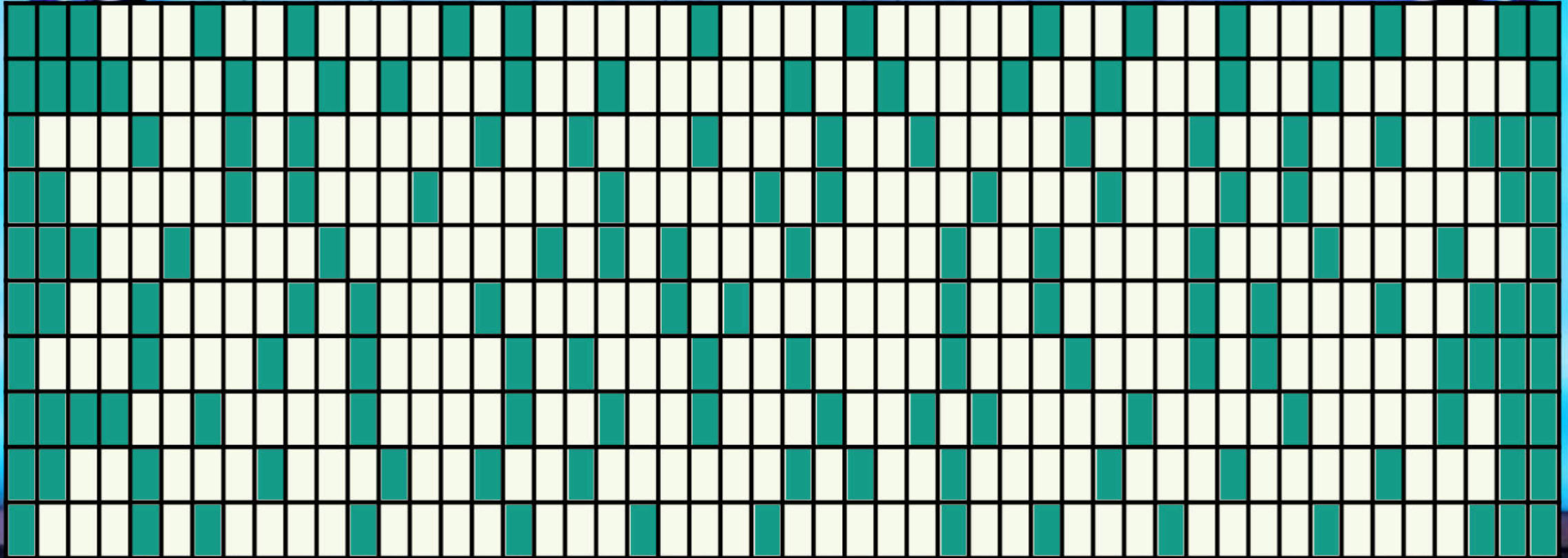
I ' M S O R R Y T O
H E A R T H A T Y O U
H A V E A R A S H O N
Y O U R B A C K

WHEEL OF
FORTUNE

GUESS THE GPT RESPONSE!

THE
WORD
FINDER

Guessing Responses is HARD!

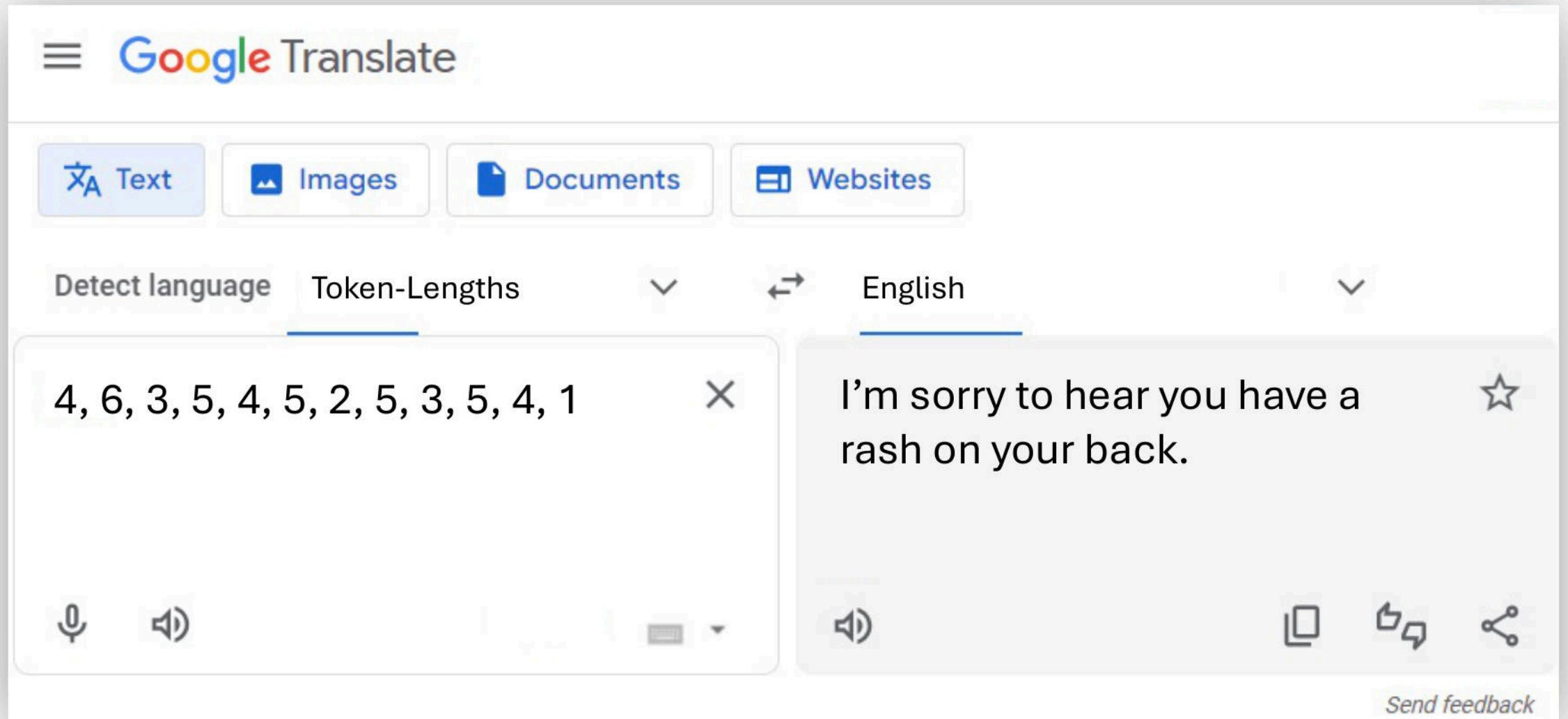


**WHEEL OF
FORTUNE**

GUESS THE GPT RESPONSE!

THE
WORD
FINDER

Our Task - Translation



The screenshot shows the Google Translate web interface. At the top, there is a hamburger menu icon and the "Google Translate" logo. Below the logo, there are four buttons: "Text" (selected), "Images", "Documents", and "Websites". Under the "Text" button, there are two tabs: "Detect language" and "Token-Lengths" (which is underlined). To the right of these tabs is a language selection dropdown currently set to "English". The main area is split into two panels. The left panel contains the text "4, 6, 3, 5, 4, 5, 2, 5, 3, 5, 4, 1" and a close button (X). The right panel contains the translated text "I'm sorry to hear you have a rash on your back." and a star icon. At the bottom right, there is a "Send feedback" link.

Google Translate

Text Images Documents Websites

Detect language Token-Lengths English

4, 6, 3, 5, 4, 5, 2, 5, 3, 5, 4, 1

I'm sorry to hear you have a rash on your back.

Send feedback

Machine Learning to the Rescue!

Why ML

- Translation tools use ML
- ML learns from examples
- Excellent at pattern recognition

Large Language Models (LLM)

- They power ChatGPT and other services
- Huge context windows

GPT-Keylogger

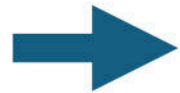


I'm sorry to hear you have
a rash on your back.



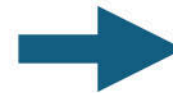
Token Lengths

4, 6, 3, 5, 4, 5,
2, 5, 3, 5, 4, 1



Attack Model

*LLM trained on
many examples*



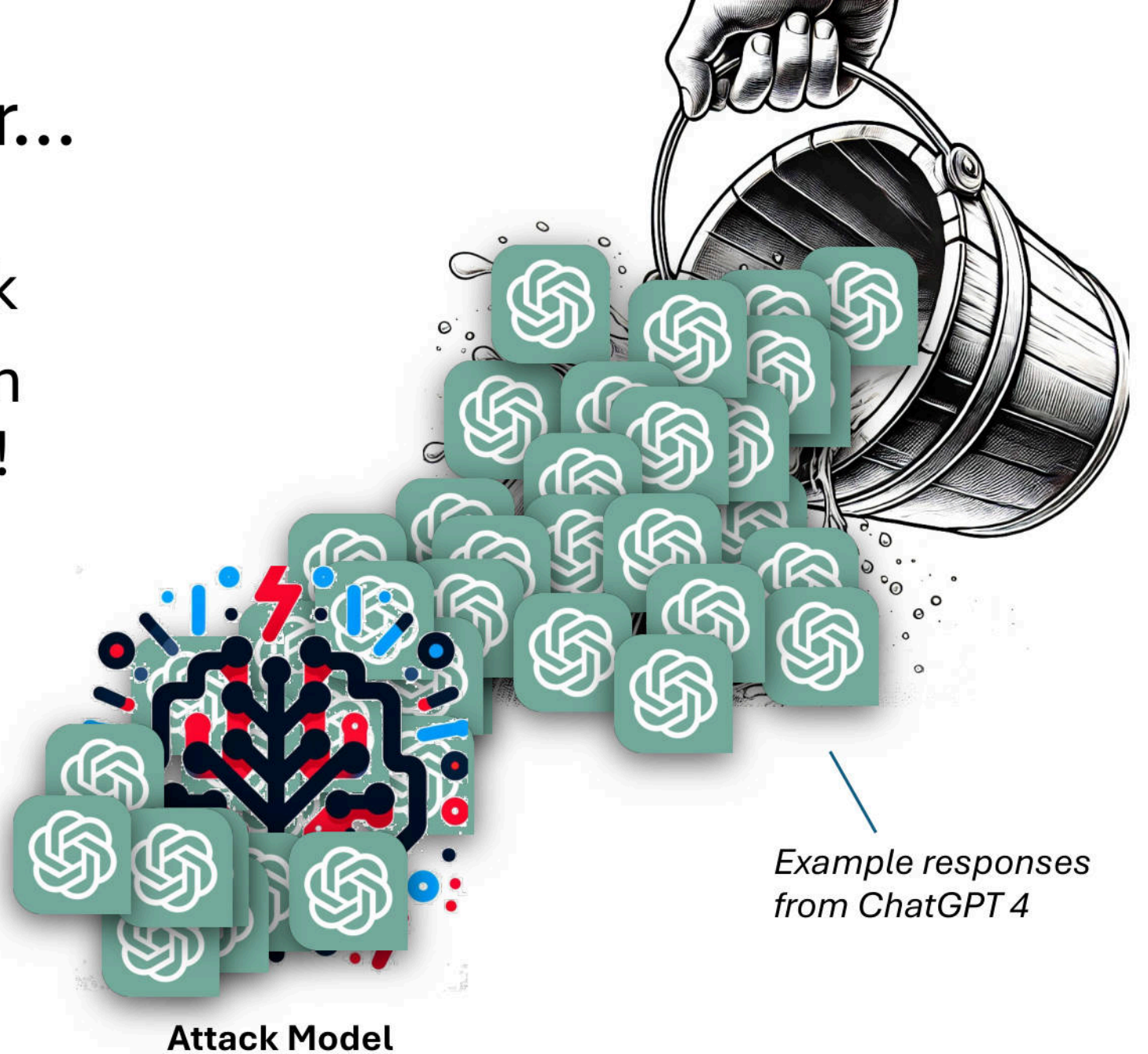
Predicted Text

I'm sorry to hear you have a
rash on your **face**.



Making it Even Better...

- Known-plaintext attack
- Fine-tune the model on the target chat service!



Attack Success Rates on **OpenAI GPT-4**



Predicting First Sentence: 55%
Predicting Entire Text: 38%

More stats to follow...

Successes – First Sentence

ϕ : 1.00 ROUGE-1: 1.00 Edit Distance: 0.00

The most common signs and symptoms of depression in young adults include:

The most common signs and symptoms of depression in young adults include:

ϕ : 0.80 ROUGE-1: 0.83 Edit Distance: 0.21

I don't have personal experience in **qualifying** for a career **option**.

I don't have personal experience in **preparing** for a career **change**,

ϕ : 0.92 ROUGE-1: 0.88 Edit Distance: 0.12

Yes, there are several **important** legal considerations that **couples** should be aware of when considering a divorce,

Yes, there are several **potential** legal considerations that **someone** should be aware of when considering a divorce.

ϕ : 0.66 ROUGE-1: 0.80 Edit Distance: 0.17

Alcohol consumption can have a harmful effect on **liver function**.

Alcohol consumption can have a harmful effect on **sleep patterns**.

Successes – First Paragraph

ϕ : 0.93 ROUGE-1: 0.75 Edit Distance: 0.26

Original: As an AI language model, I don't have personal experience or emotions, but below are some strategies that can help in developing a self-compassionate attitude:

1. Practice mindfulness - Mindfulness can help you identify and observe your thoughts and emotions without judging or reacting to them. This can help you develop a more compassionate attitude towards yourself.
2. Treat yourself with kindness - In stressful situations, treat yourself the way you would treat a friend or loved one. Do something kind for yourself, such as taking a break, self-care activities, or giving yourself a compliment.
3. Change your self-talk - Instead of being self-critical or negative, try to reframe your self-talk in a positive and supportive manner.
4. Accept your imperfections - Acknowledge that everyone makes mistakes and has flaws, including yourself. Learn to accept yourself for who you are, imperfections and all.
5. Be patient with yourself - Give yourself time to process and bounce back from stressful situations. Remember that healing and growth take time.
6. Seek support - Reach out to friends, family, or a mental health professional for support, guidance,

Prediction: As an AI language model, I don't have personal experience or emotions, but below are some strategies that can help in developing a self-compassion attitude:

1. Practice mindfulness - Mindfulness can help you identify and process your emotions and thoughts without judging or reacting to them. This can help you develop a more compassionate attitude towards yourself.
4. Treat yourself with kindness - By embracing compassion, treat yourself the way you would treat a friend or loved one. By accepting your own mistakes, such as taking a break, resuming activities, or simply avoiding a temptation.
4. Change your self-talk - Instead of going into a battle or argument, try to reframe your self-talk in a positive and supportive manner.
3. Accept your imperfections - Acknowledge that everyone makes mistakes and has flaws. Challenge yourself, learn to accept yourself for who you are, treat oneself kindly.
5. Be patient with yourself - Your personal life is complex and unique with many different challenges. Practice self-control and manage your time.
6. Seek support - Reach out to friends, family, or a mental health professional for support, guidance,

Fails – First Sentence

ϕ : 0.44 ROUGE-1: 0.64 Edit Distance: 0.36

I would suggest the following strategies for **team leaders** to balance the needs of **individual team members with the needs of the team as a whole**:

I would suggest the following strategies for **film studios** to balance the needs of **production with staying true to the story of a movie**:

ϕ : 0.12 ROUGE-1: 0.00 Edit Distance: 0.78

Cider and other apple-based beverages hold great significance in Normandy's culture and cuisine.

Coral bleaching occurs after organisms lose their reproduction or grow out a suitable new habitat.

How common is this vulnerability?

Before disclosure Feb '24



copy.ai



CKSOURCE™

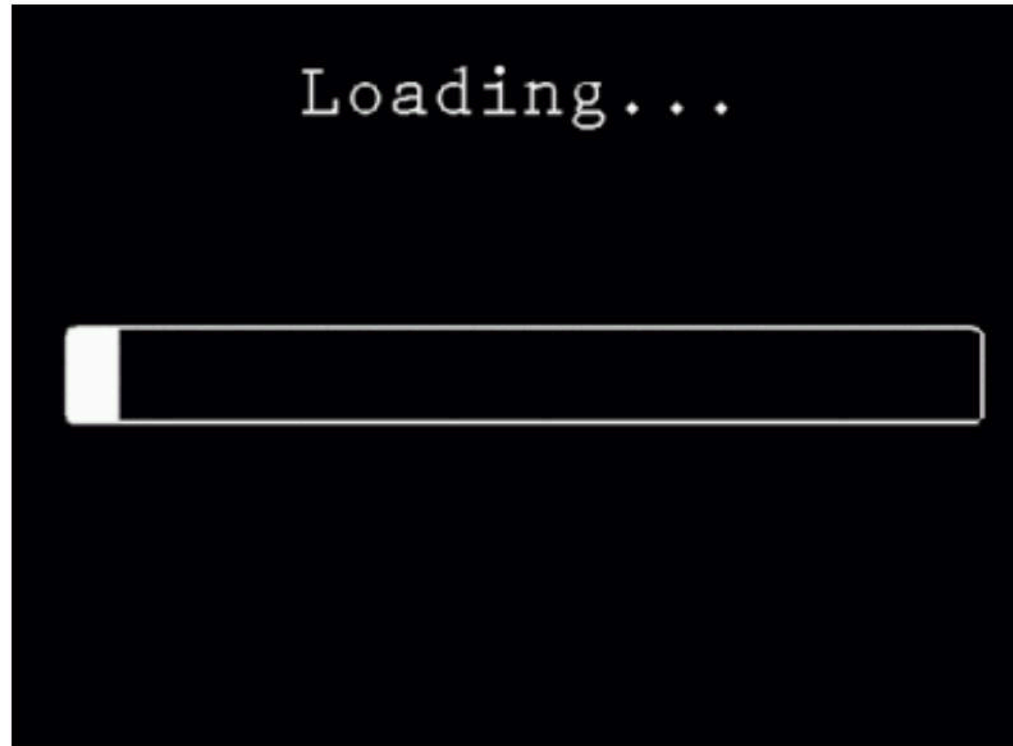


Ah shit, here we go again.

Why is this vulnerability overlooked?

1. Token lengths $\neq$ word lengths
2. Problem considered too hard for adversary
 - Guessing with no context
 - Long word length sequences
3. AI Gold Rush

Okay, but why stream the tokens?



Recap so far...

Vulnerability

- Discovery of the token-length side channel
- Common in LLM services

Exploitation

- GPT-Keylogger
 - A custom AI model that can **use** the side channel
 - Known-plaintext attack

The Rest of this Talk

Deep Dive

Finding the
Side Channel



Token-Length
Extraction



Exploitation
with AI



Demos &
Defences



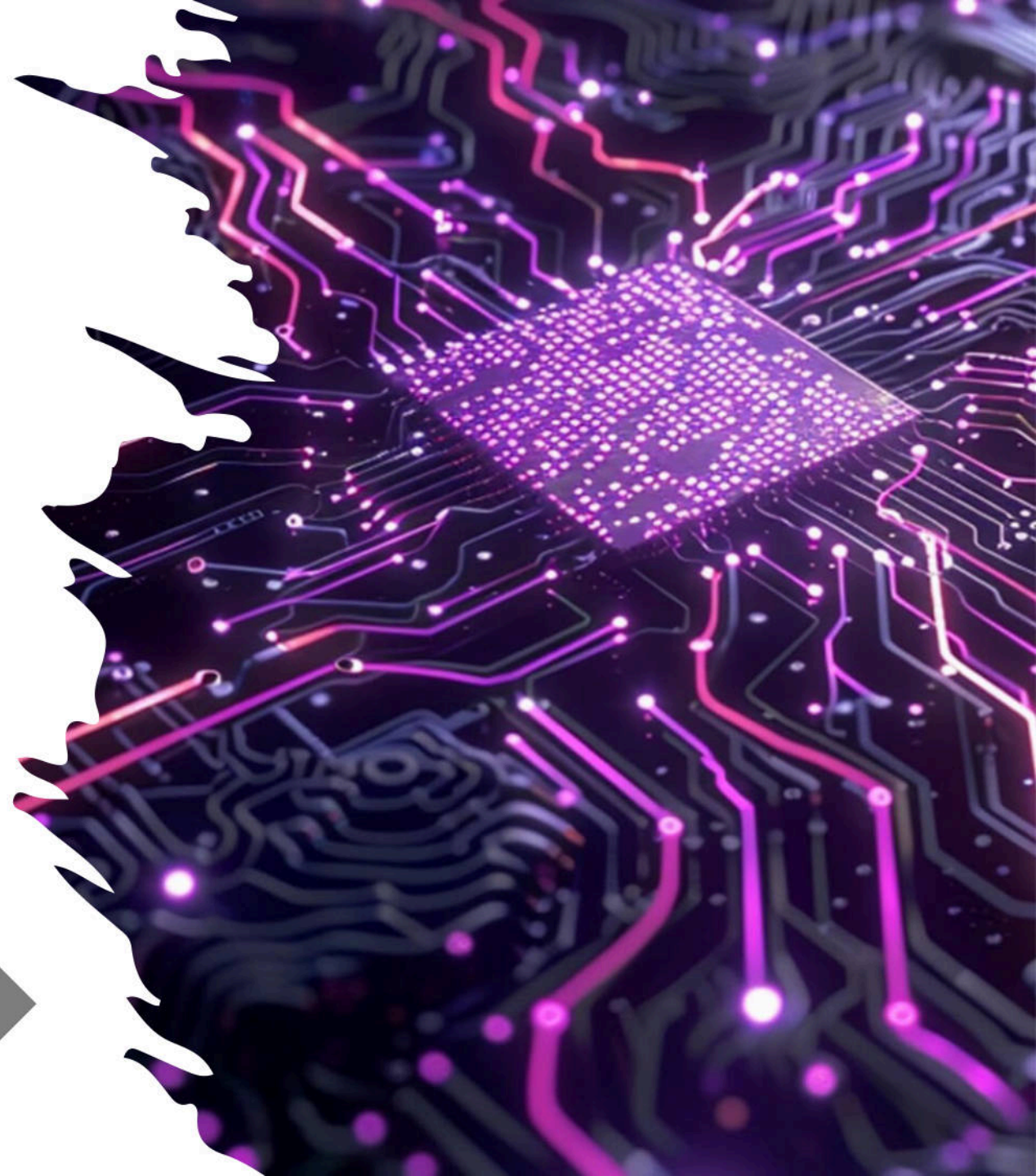
Finding the Side Channel

Finding the
Side Channel

Token-Length
Extraction

Exploitation
with AI

Demos &
Defences



Traffic Capture

The attacker needs to be able to sniff the victims incoming traffic.

- Same local network (e.g. Café, Library, etc...)
- ISP
- ...



Filtering the Traffic

Objective: Find Response Traffic

- Filter by IP address
 - OSINT
 - Connect as client
- Filter by protocol



Protocols

- Different Protocols are used to stream tokens
- Depends on the vendor

Protocol	Vendor(s)
QUIC	ChatGPT
WebSocket	Microsoft CoPilot, Preplexity
HTTP2	Claude, Writesonic, Notion AI, ClickUp AI, Text Cortex AI
TCP	Quora Poe, OpenAI API, CloudFlare AI API

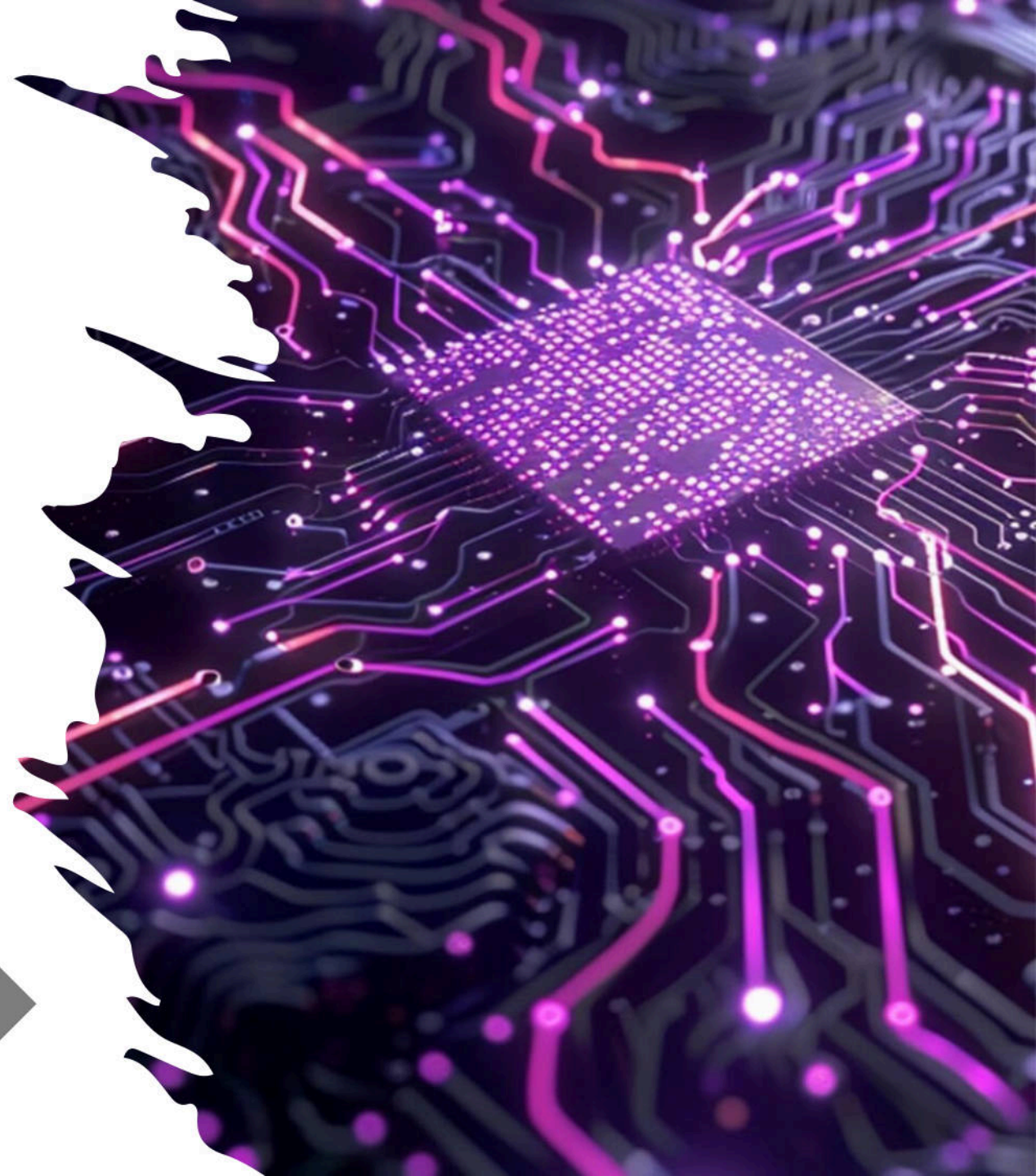
Token-Length Extraction

Finding the
Side Channel

Token-Length
Extraction

Exploitation
with AI

Demos &
Defences



From Traffic to Token-length Sequence

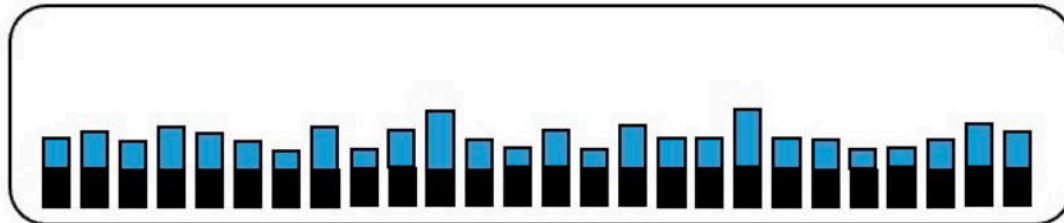
Objective: find the token-lengths

- Traffic is encrypted
- But, we can see bandwidth patterns

How Tokens are Sent

We have observed 2 ways vendors send the next token.

Base + Next Token



ClickUp



CKEditor™



CKSOURCE™

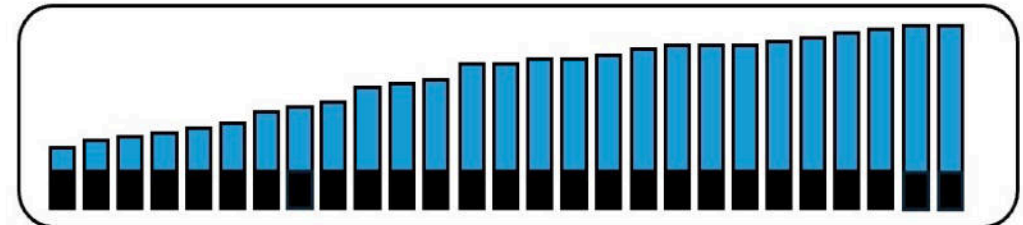


Writesonic



text.cortex

Token Accumulation



copy.ai

Quora

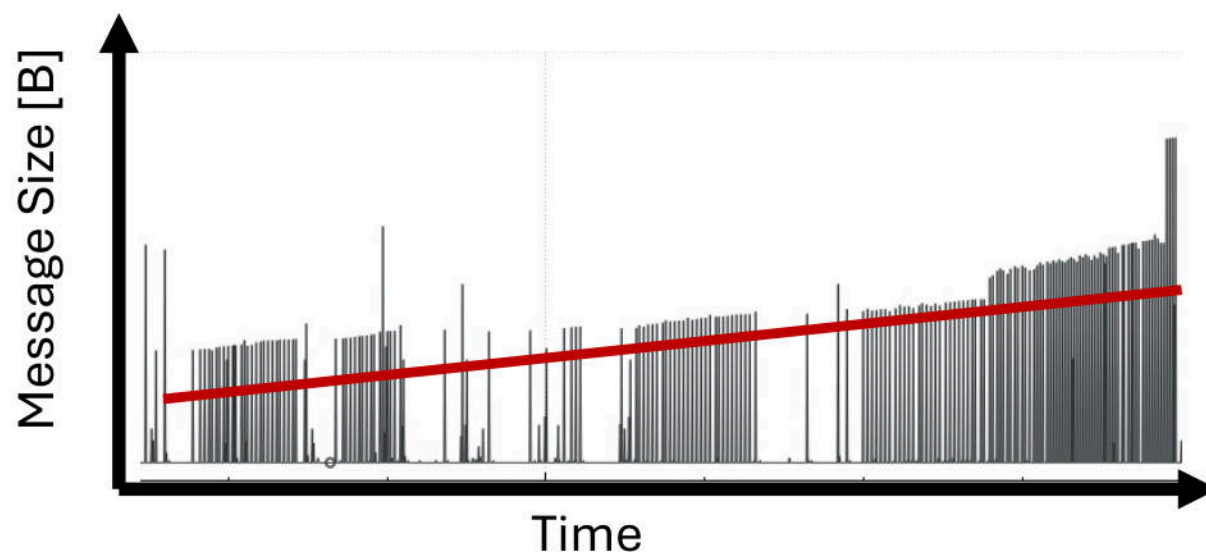


perplexity

Identifying the Messages from a Response

Finding the start and end of a response

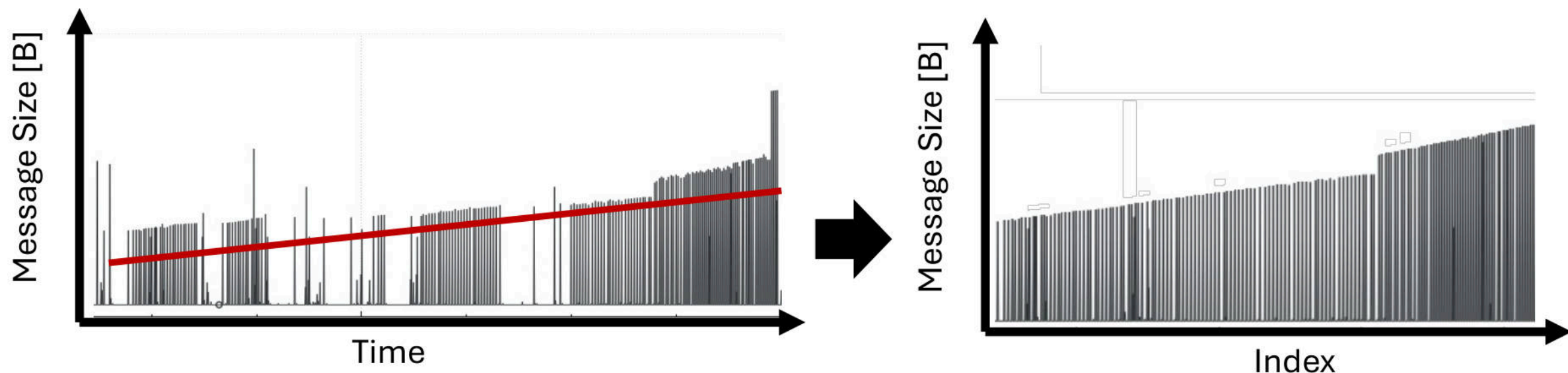
💡 Remember: every vendor is different.



* Traffic is **encrypted**, so we must find message tokens based on bandwidth patterns alone

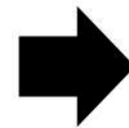
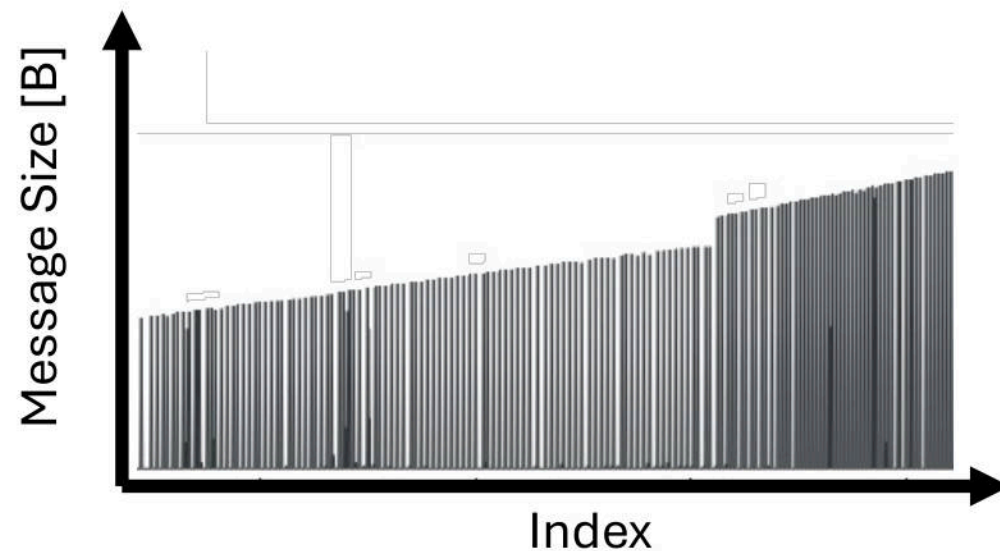
Cleaning it Up

- Some messages **do not** contain tokens
 - Handshakes, control, ...
- We remove these by following the monotonic trend
- ...there is probably a better way



Token Length Extraction

To get the token lengths, we compute the delta between messages



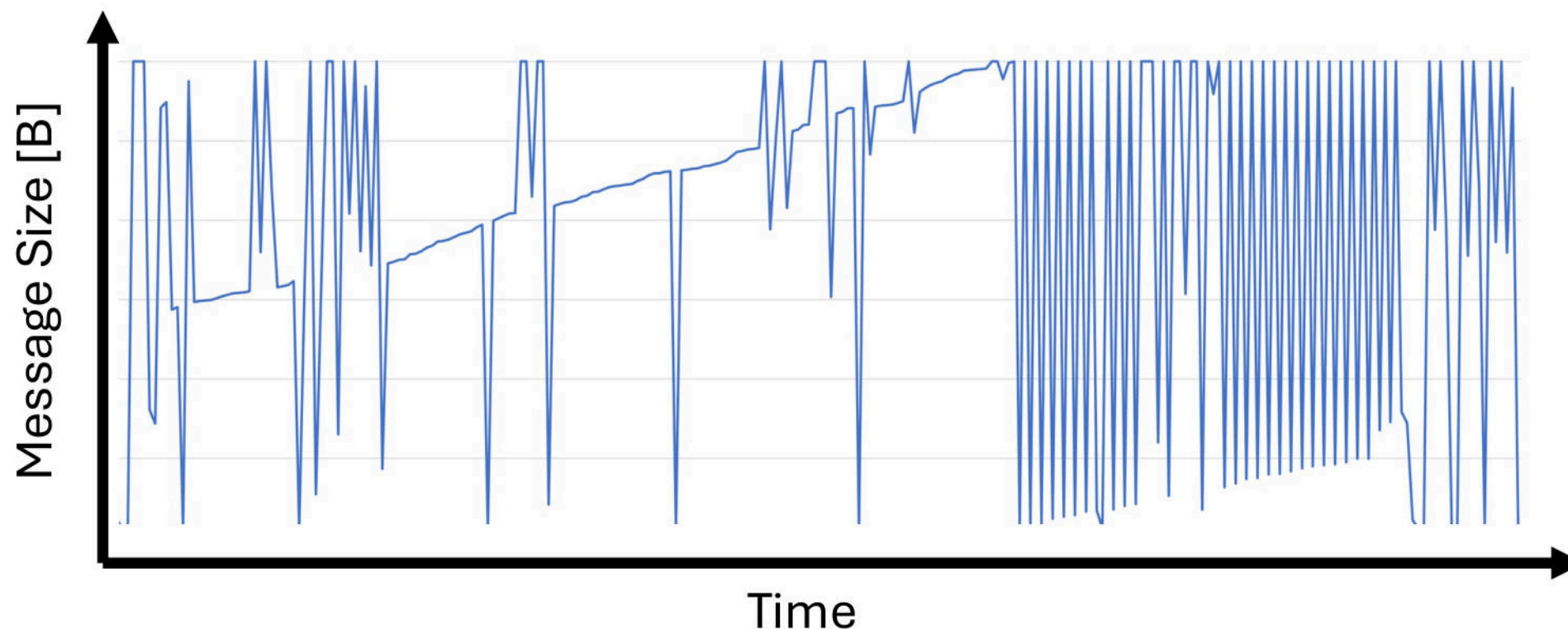
4, 6, 3, 5, 4, 5, 2, 5, 3, 5, 4, 1, ...

Real World Challenges

- Message Size Limits
- Message Buffering -> Message Grouping

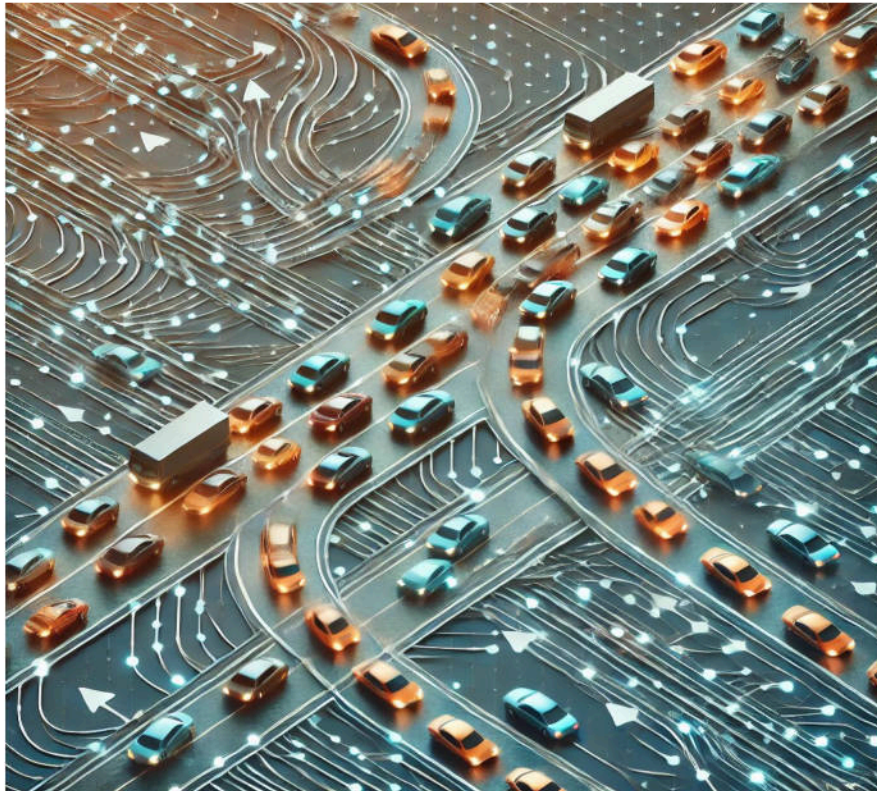
Message Size Limits

- Maximum transmission unit, QUIC has even lower max message size
- Results in a saw-tooth bandwidth pattern

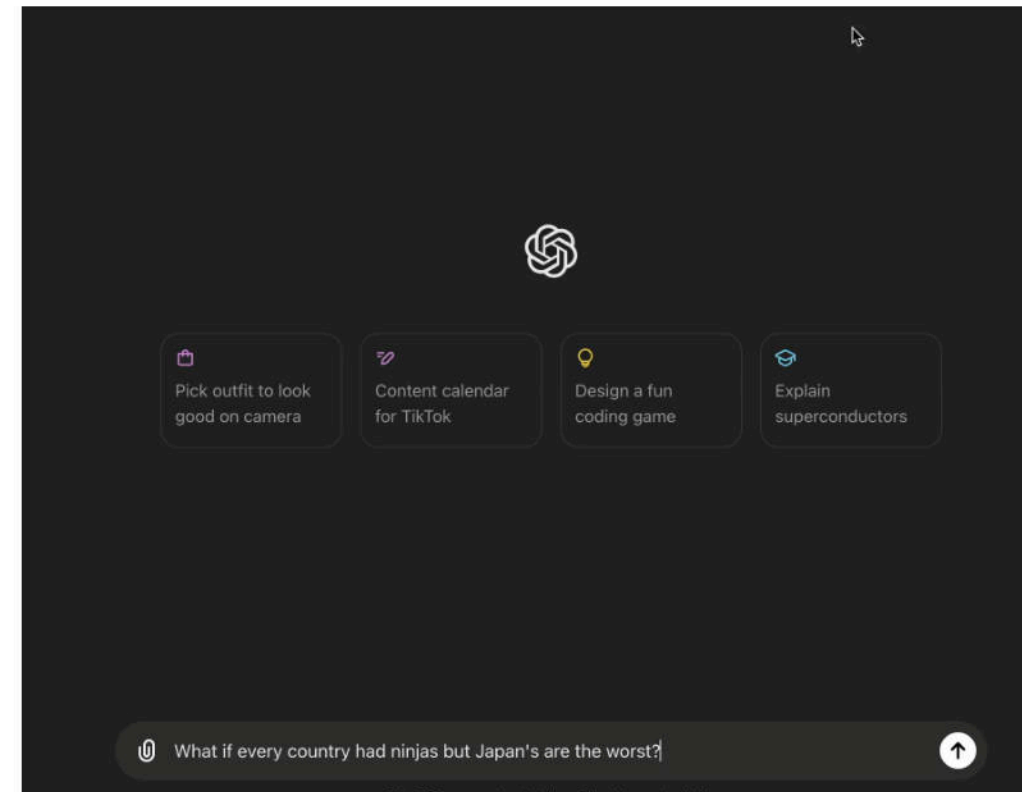


Message Buffering

Busy Network

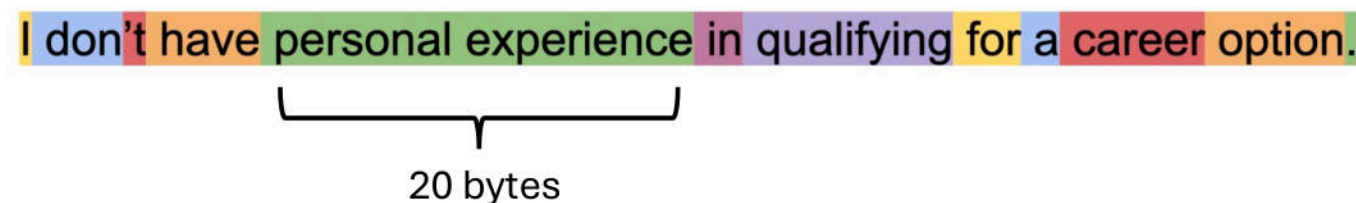


More Buffering

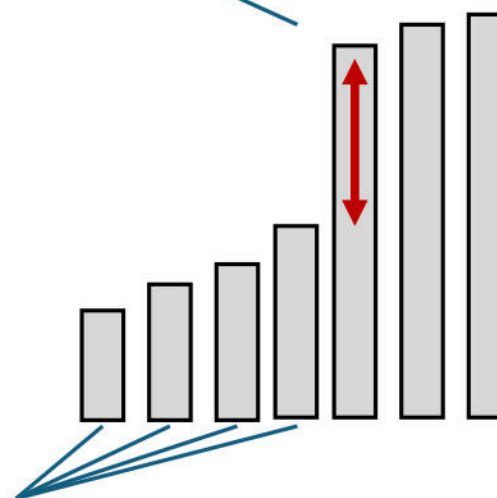


Message Buffering

- Our main target was **OpenAI's ChatGPT**
- A challenge with QUIC
 - Messages can be assembled on client side
- Sometimes we get grouped messages



This message has
several tokens



These messages
have only one

How did we figure this all out?

Reverse Engineering

Step 1: Connect as legitimate client

Step 2: Grab SSL key

Step 3: Plug it into Wireshark



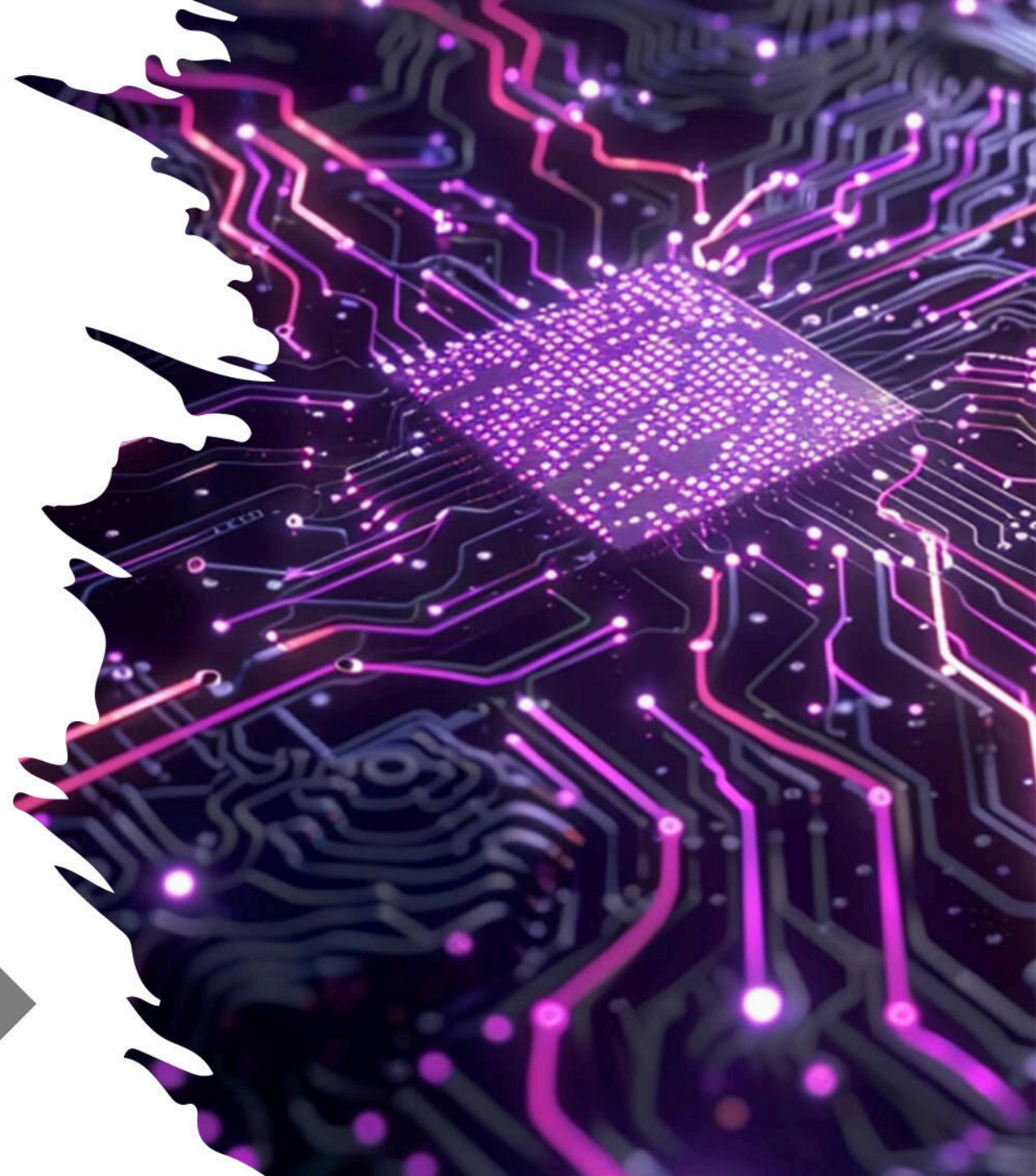
Exploitation with AI

Finding the
Side Channel

Token-Length
Extraction

Exploitation
with AI

Demos &
Defences



How to Train your ML Model



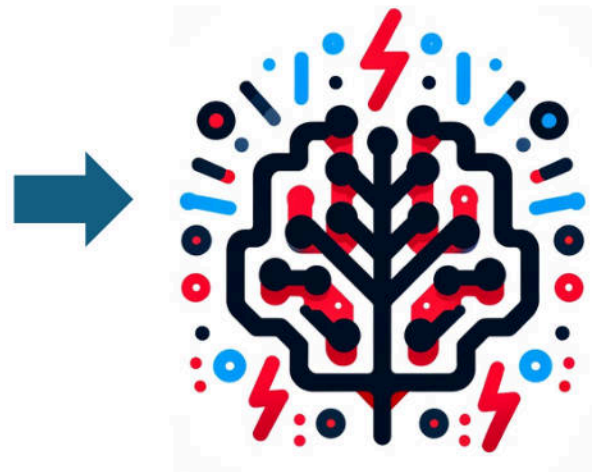
The Procedure

Create Dataset

The best way to save money...	3, 5, 4, 3, 5, 6 ...
Planning a trip to Europe...	8, 2, 5, 3, 7 ...
Finding a job you love requires...	7, 2, 4, 4, 5, 9 ...
⋮	⋮

Train Model

3, 5, 4, 3, 5, 6 ...
8, 2, 5, 3, 7 ...
7, 2, 4, 4, 5, 9 ...
⋮



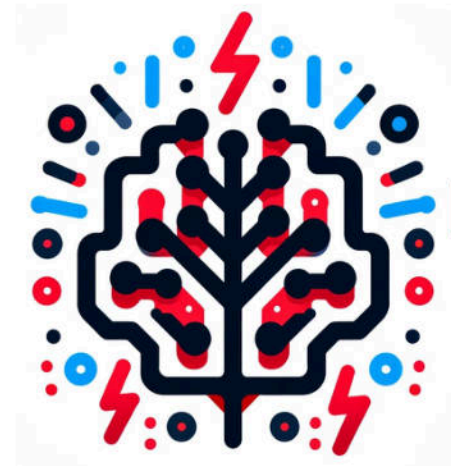
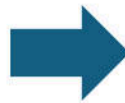
The best way to save money...
<u>Organizing</u> a trip to Europe...
<u>Securing</u> a job you love requires...
⋮

The Procedure

Use The Model



3, 4, 5, 2, 5, 5...



Generate Options & Rank them



If you have a rash then...
If you have a **pain** then...
⋮
If **one cuts** a **pear five**...
⋮

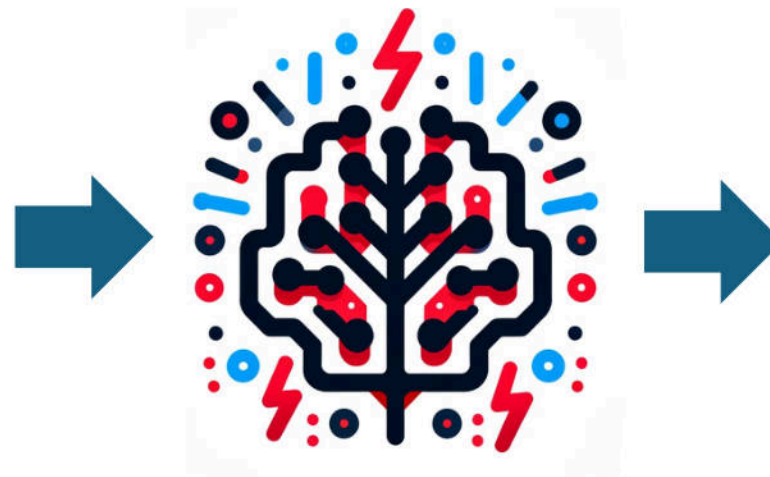
And...

Dealing with a rash on your back involves several steps:

1. Identify the cause: Determine if the rash is due to an allergy, irritation, infection, or other condition.
2. Keep the area clean: Gently wash the rash with mild soap and water.
3. Apply a soothing lotion: Use calamine lotion or an anti-itch cream to relieve discomfort.
4. Avoid scratching: Scratching can worsen the rash and lead to infection.
5. Wear loose clothing: Prevent further irritation by wearing loose, breathable fabrics.
6. ...

6, 5, 2, 5, 3, 5, 5, 7, 8,
6, 1, 1, 8, 4, 6, 8, 3, 7,
3, 2, 1, 1, 5, 8, 4, 3 ...

**Token-length
Sequence**



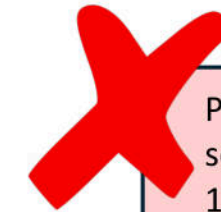
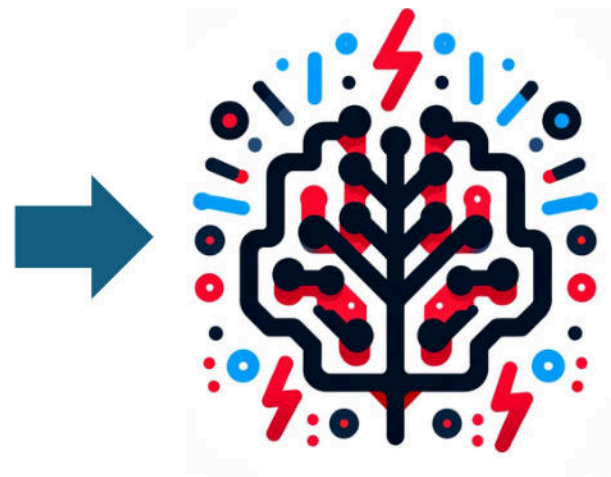
And... didn't work :(

Dealing with a rash on your back involves several steps:

1. Identify the cause: Determine if the rash is due to an allergy, irritation, infection, or other condition.
2. Keep the area clean: Gently wash the rash with mild soap and water.
3. Apply a soothing lotion: Use calamine lotion or an anti-itch cream to relieve discomfort.
4. Avoid scratching: Scratching can worsen the rash and lead to infection.
5. Wear loose clothing: Prevent further irritation by wearing loose, breathable fabrics.
6. ...

6, 5, 2, 5, 3, 5, 5, 7, 8,
6, 1, 1, 8, 4, 6, 8, 3, 7,
3, 2, 1, 1, 5, 8, 4, 3 ...

**Token-length
Sequence**



Preparing for a job interview involves several steps:

1. Research the company: Learn about their mission, values, and recent news.
2. Review the job description: Understand the skills and qualifications required.
3. Practice common questions: Rehearse answers to typical interview questions.
4. Plan your outfit: Choose professional and appropriate attire.
5. Bring necessary documents: Have copies of your resume and references ready.
6. ...

Eureka!

Supporting a friend with depression involves being present and understanding.

Listen without judgment and offer a safe space for them to express their feelings. Encourage them to seek professional help from a therapist or counselor. Avoid giving unsolicited advice or trying to "fix" their problems. Instead, reassure them of your support and remind them that they are not alone. Engage in activities they enjoy, and be patient as they navigate their journey to...

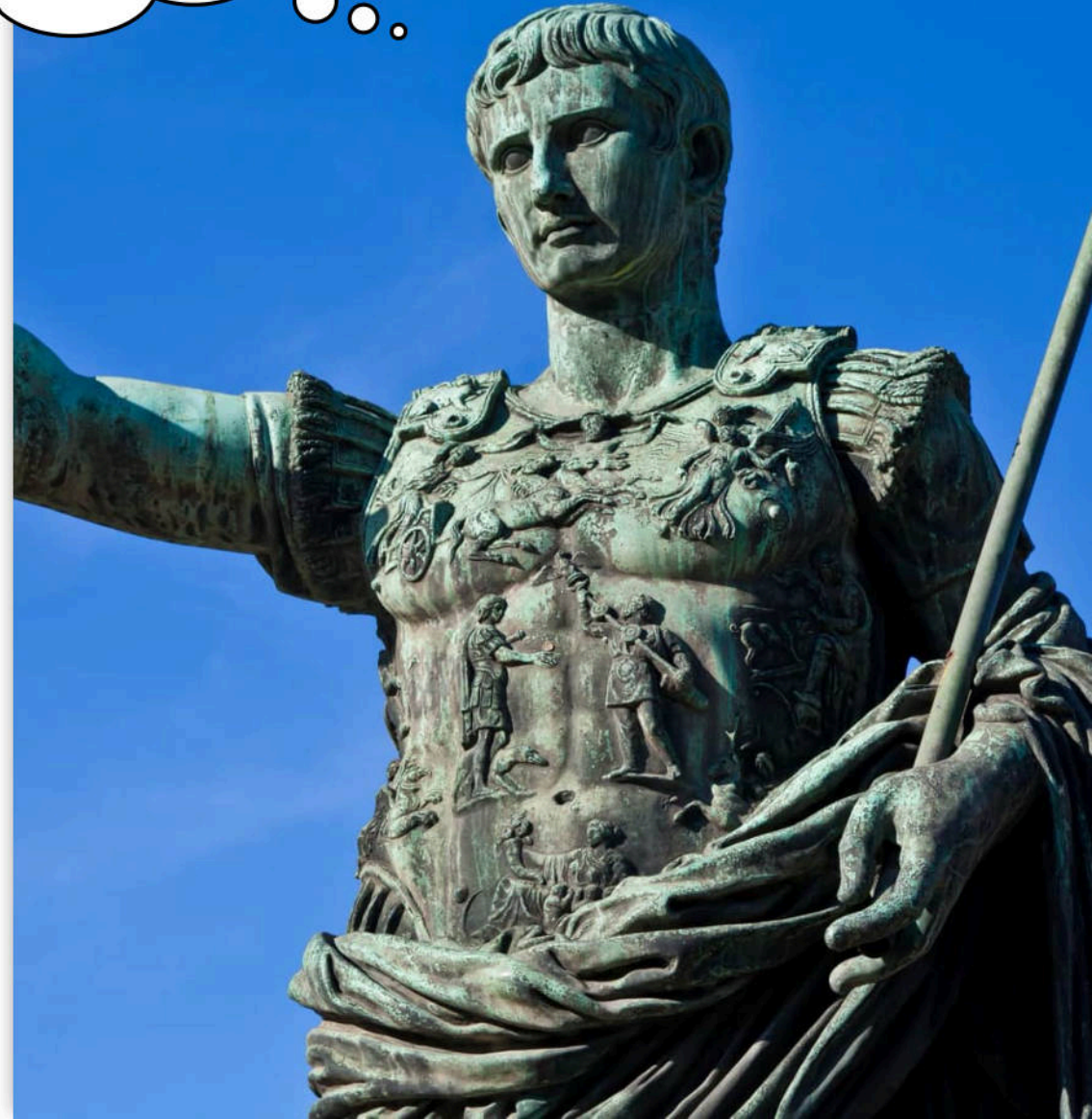
If you suspect your child is being bullied, the first step is to talk to them openly and calmly. Create a safe environment for them to share their experiences without fear of judgment. Document any incidents they describe, noting dates, times, and details. Contact the school to discuss your concerns with teachers, counselors, or administrators. Advocate for your child by requesting a plan of action to address the bullying. Encourage your child to develop coping strategies...

Discussing substance abuse with a loved one requires sensitivity and care. Choose a private and comfortable setting to have the conversation. Approach the topic with empathy and concern rather than judgment or accusation. Use "I" statements to express your feelings and observations, such as "I've noticed you've been drinking more lately, and I'm really worried about you." Encourage them to share their perspective and listen actively without interrupting. Offer...

Divide and Conquer

Discussing substance abuse with a loved one requires sensitivity and care. Choose a private and comfortable setting to have the conversation. Approach the topic with empathy and concern rather than judgment or accusation. Use "I" statements to express your feelings and observations, such as "I've noticed you've been drinking more lately, and I'm really worried about you." Encourage them to share their perspective and listen actively without interrupting.

Who thought
I'd end up at
DEF CON



Divide and Conquer

Discussing substance abuse with a loved one requires sensitivity and care.

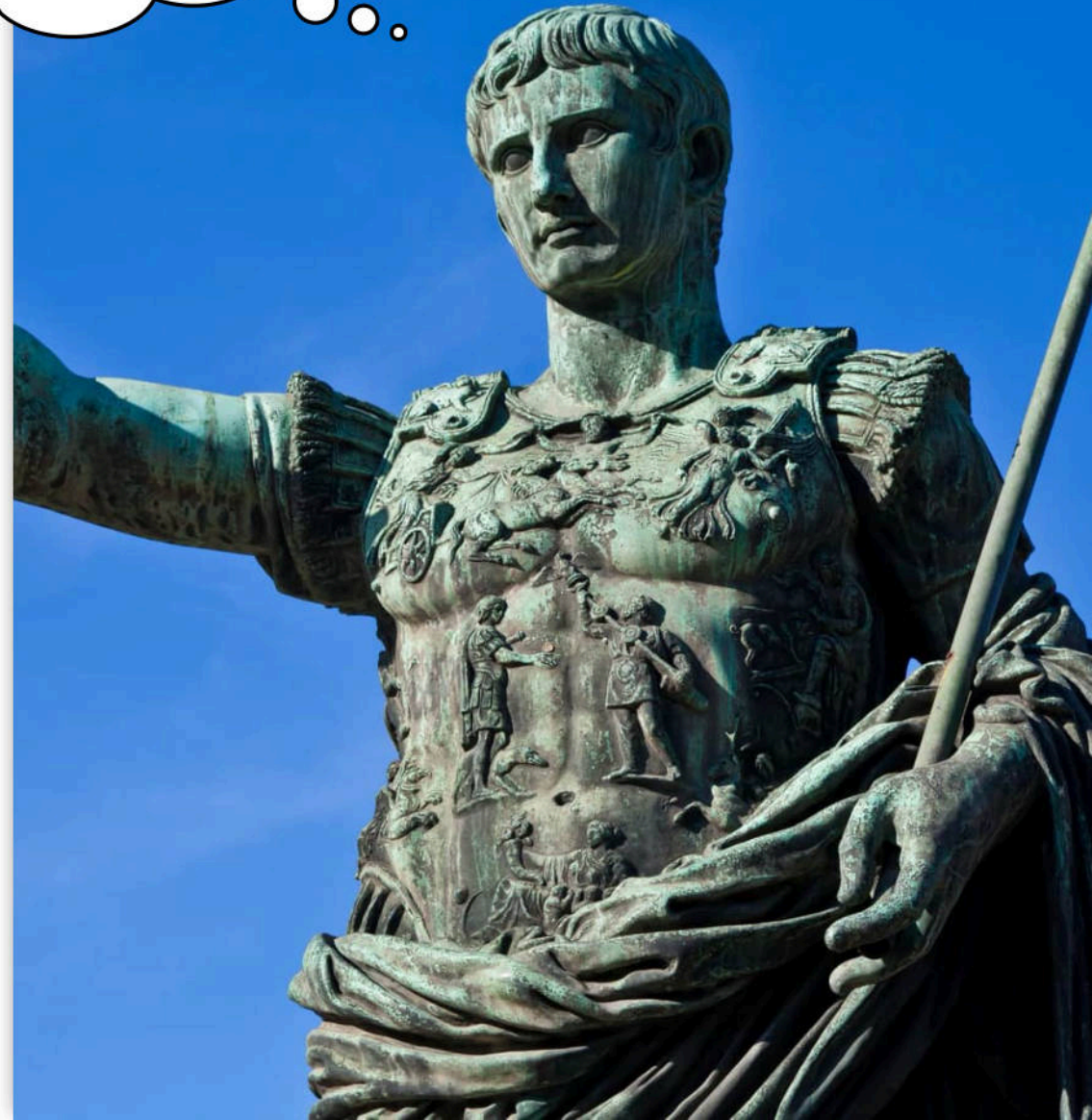
Choose a private and comfortable setting to have the conversation.

Approach the topic with empathy and concern rather than judgment or accusation.

Use "I" statements to express your feelings and observations, such as "I've noticed you've been drinking more lately, and I'm really worried about you."

Encourage them to share their perspective and listen actively without interrupting.

Who thought
I'd end up at
DEF CON



How do we find sentences?

Token-length sequence (3 sentences):

1 5 5 8 6 5 5 1 6 3 3 1 4 5 5 3 5 5 1 5 1 4 7 4 2 7 1

Most Tokenizers make punctuation marks (. , :) separate tokens

⇒ meaning they are sent as a separate message

⇒ meaning their length is 1

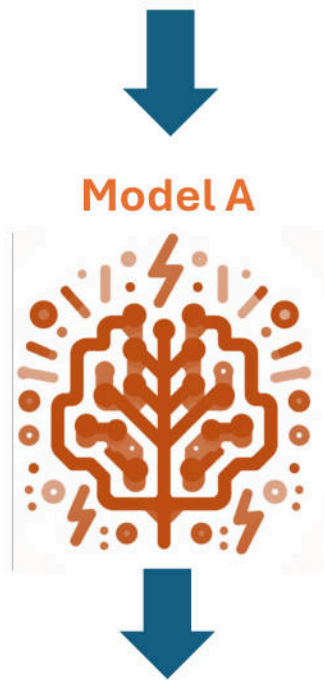
⇒ Too short, probably a comma...

1 5 5 8 6 5 5 **1** 6 3 3 **1** 4 5 5 3 5 5 **1** 5 **1** 4 7 4 2 7 **1**

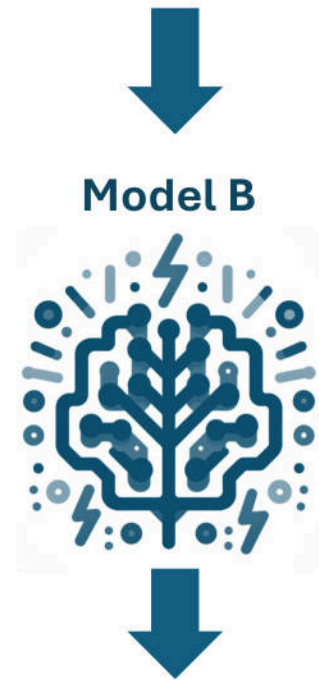
The Full Model

Parse Token-length Sequence

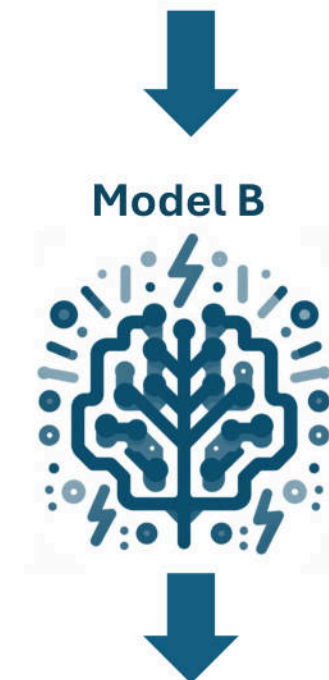
1 5 5 8 6 5 5 **1** 6 3 3 **1** 4 5 5 3 5 5 **1** 5 **1** 4 7 4 2 7 **1**



I need more details
about your rash.



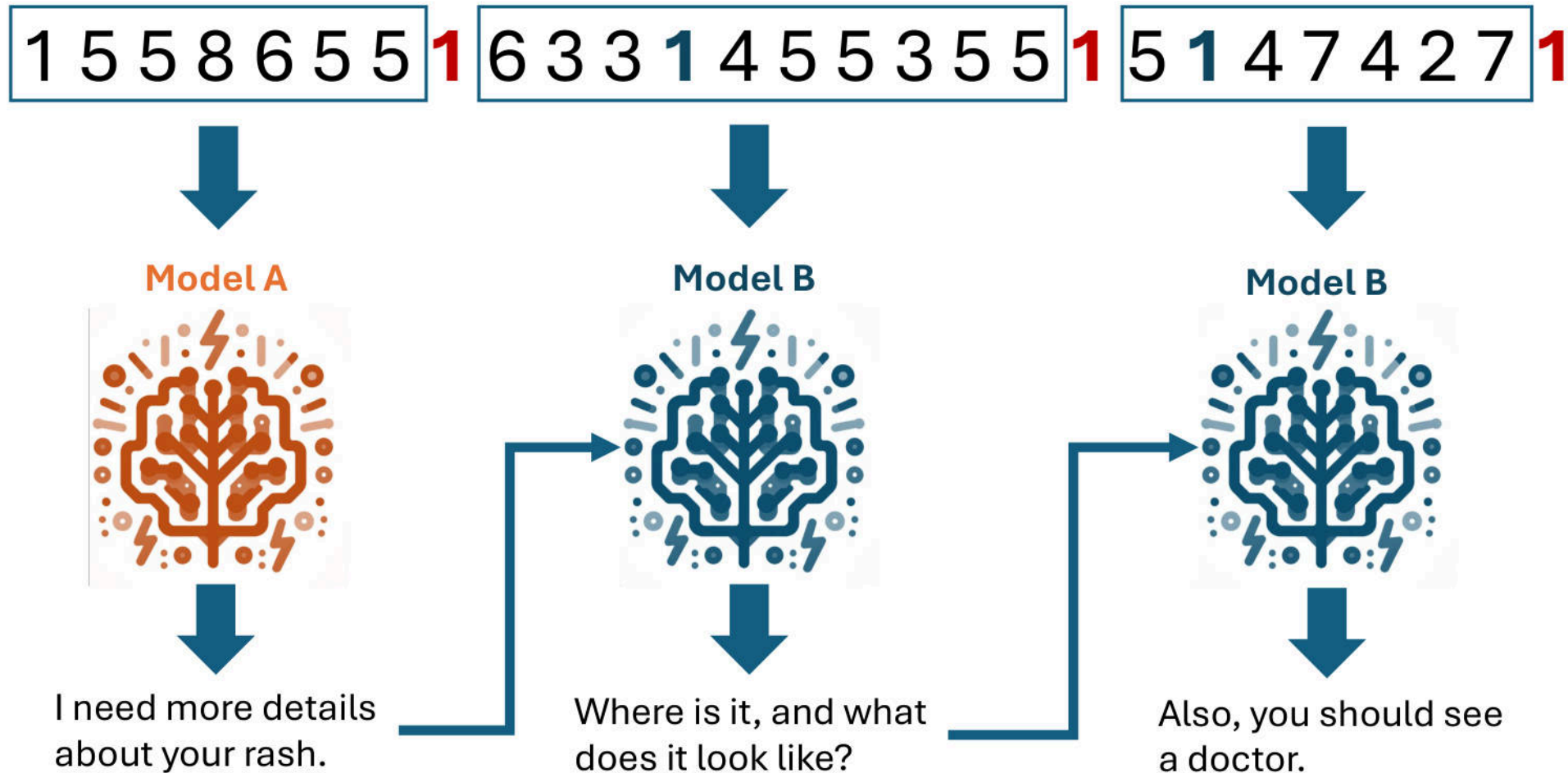
Where is it, and what
does it look like?



Also, you should see
a doctor.

The Full Model – more context

Parse Token-length Sequence



Known-plaintext Attack

AI Assistants have unique speaking style

1. Predictable Style

Mannerism, structure, ...

2. Low Response Diversity

Prompt variants lead to same response

...we can use this to our advantage

Known-plaintext Attack

Predictable Style

Warnings

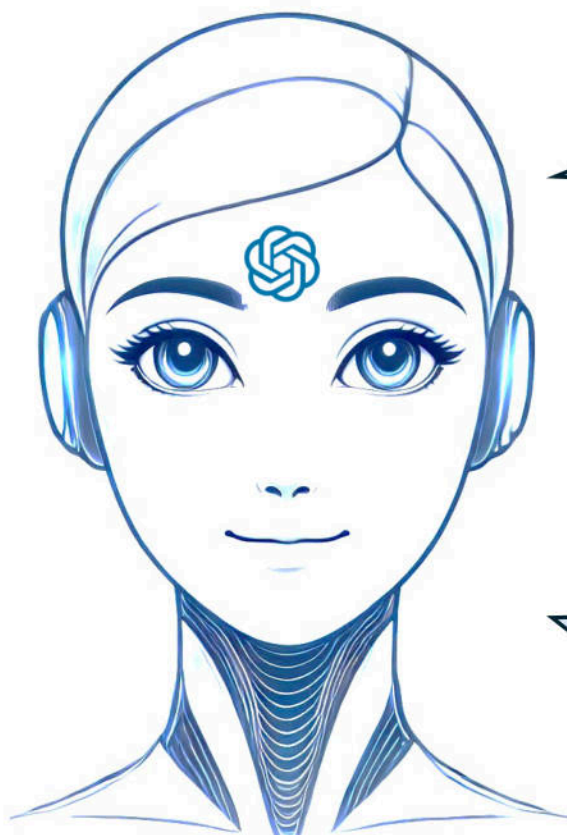
I don't have personal beliefs or interests, but..

Unique Phrases

"Yellowstone National Park"

Structure

- Apples\n
- Milk\n
- ...



Templates

Certainly! Here's a simple and classic recipe for...

AI Safeguards

I'm sorry, but I can't assist with creating content that could be used for phishing

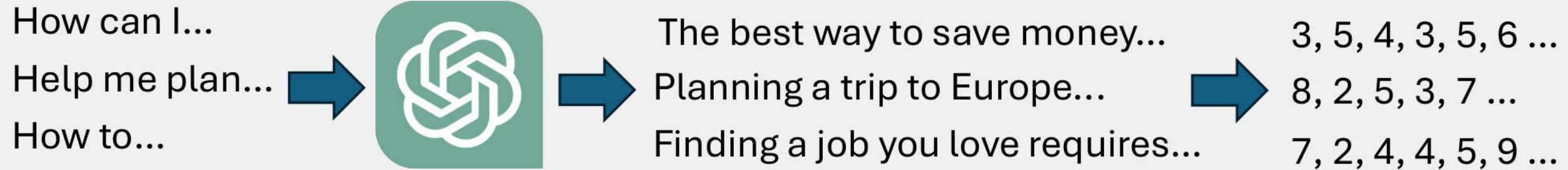
Known-plaintext Attack

Low Response Diversity

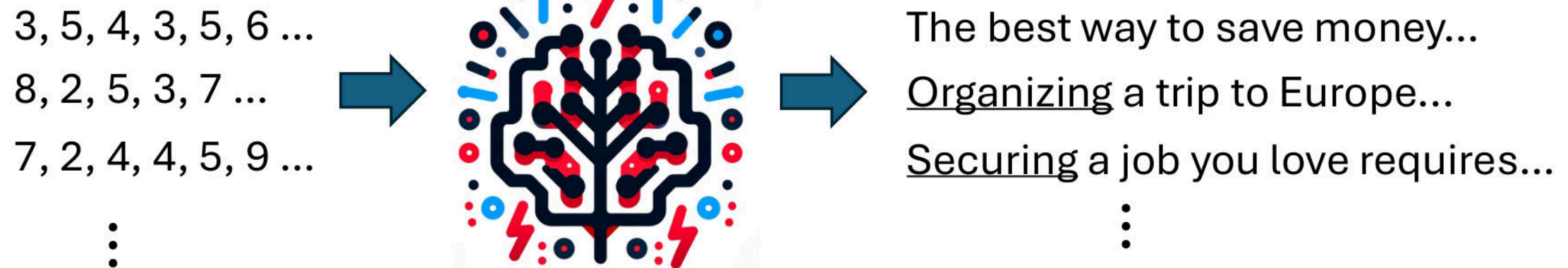


The Procedure – *known-plaintext*

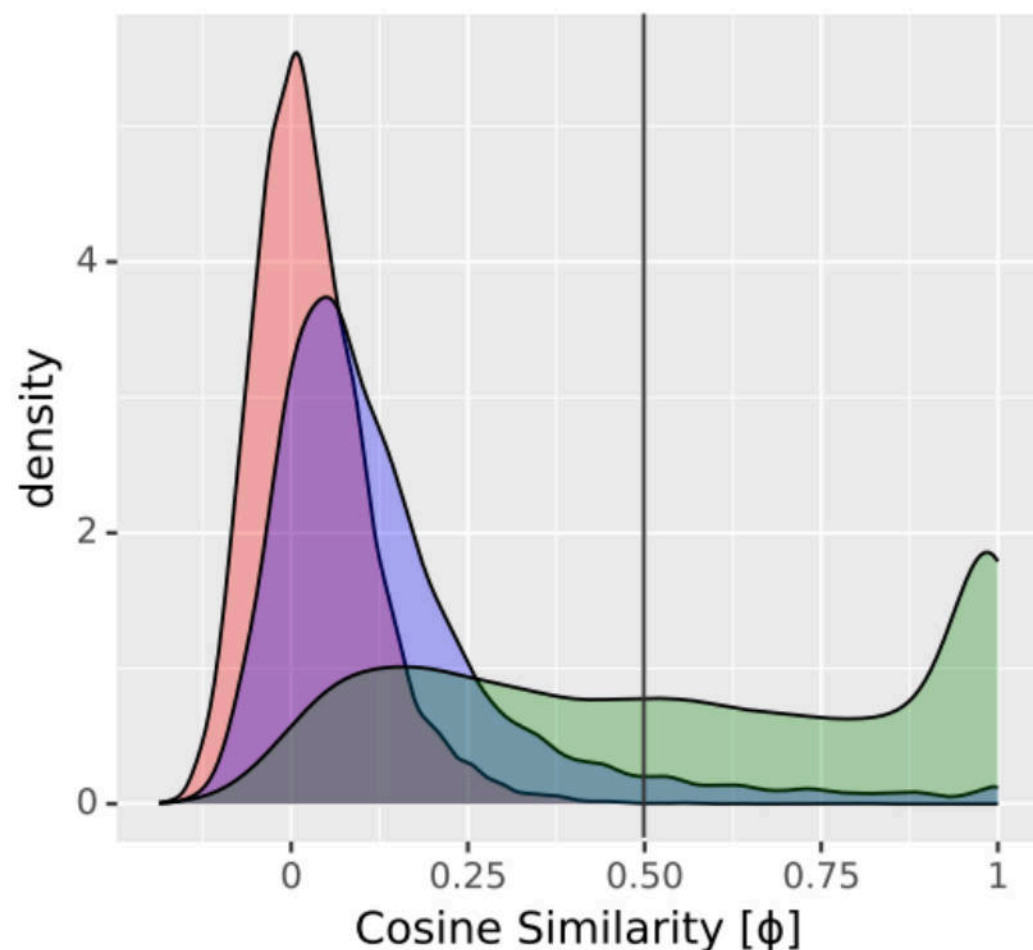
Create Dataset



Train Model



Huge Boost in Performance!

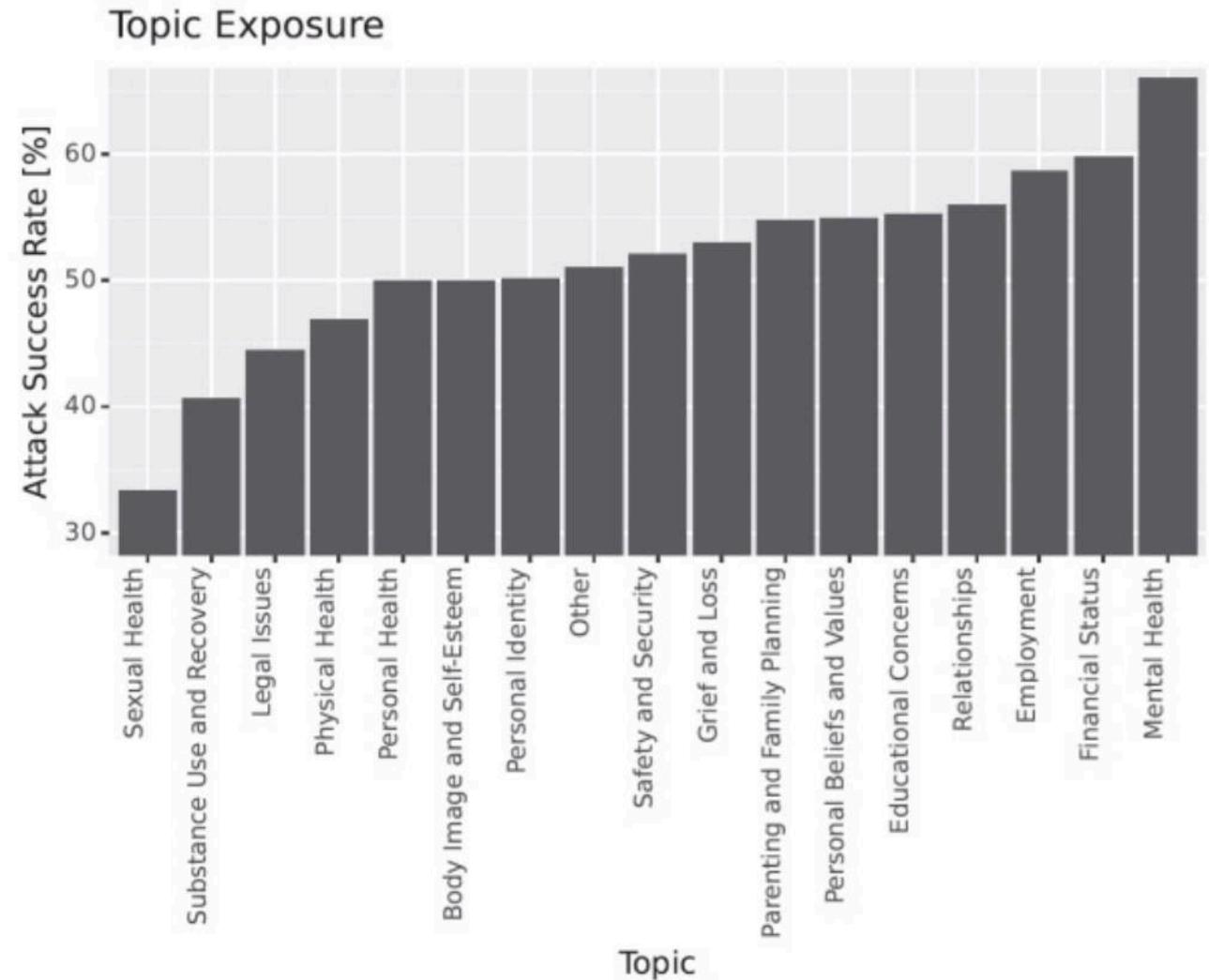


Model

- Baseline:** Asking GPT4 to solve the token-sequence
- GPT-Keylogger:** Trained on random text (C4)
- GPT-Keylogger:** Trained on GPT4 responses (UltraChat)

Performance Overall

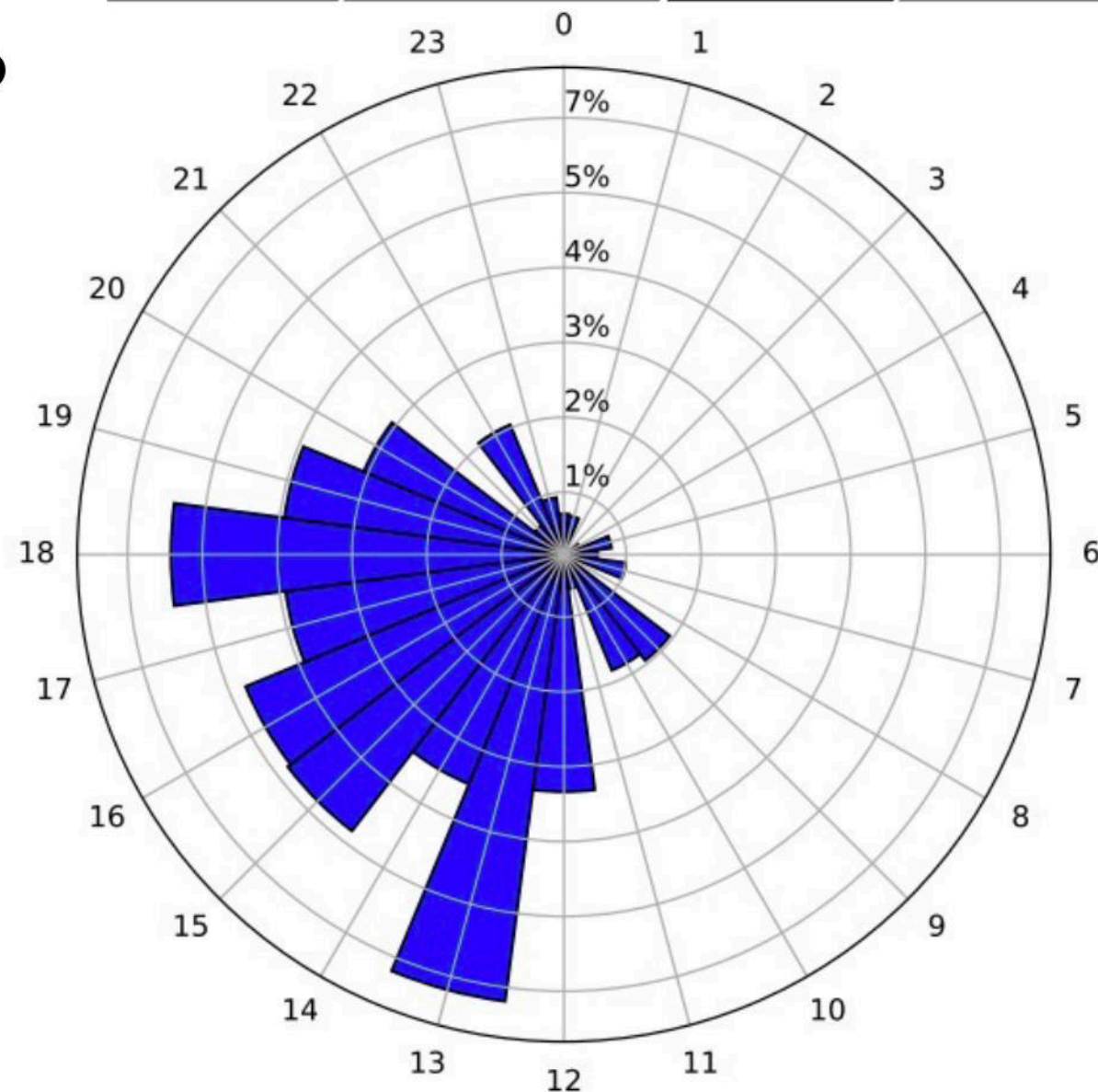
Segments	ASR
1	54.52
2	37.93
3	34.63
5	34.90
10	37.50
20	37.04
all	37.97



What About Grouping?

24 hours experiment to capture responses from OpenAI (QUIC)

Noise is measured as the percentage of buffered packets.



Percentage of Grouped Tokens for each hour

What About Grouping?

Solution: group tokens in the training set at random

Vendor	Model	Service	Topic Inference	Text Inference
OpenAI	GPT-4	Browser	35.5	13.7
OpenAI	GPT-4	GPT's	50.2	22.9
OpenAI	GPT-4	API	17.7	5.6
Microsoft	CoPilot	Browser	30.1	5.9

* We noticed the GPT's had less network interference than regular ChatGPT

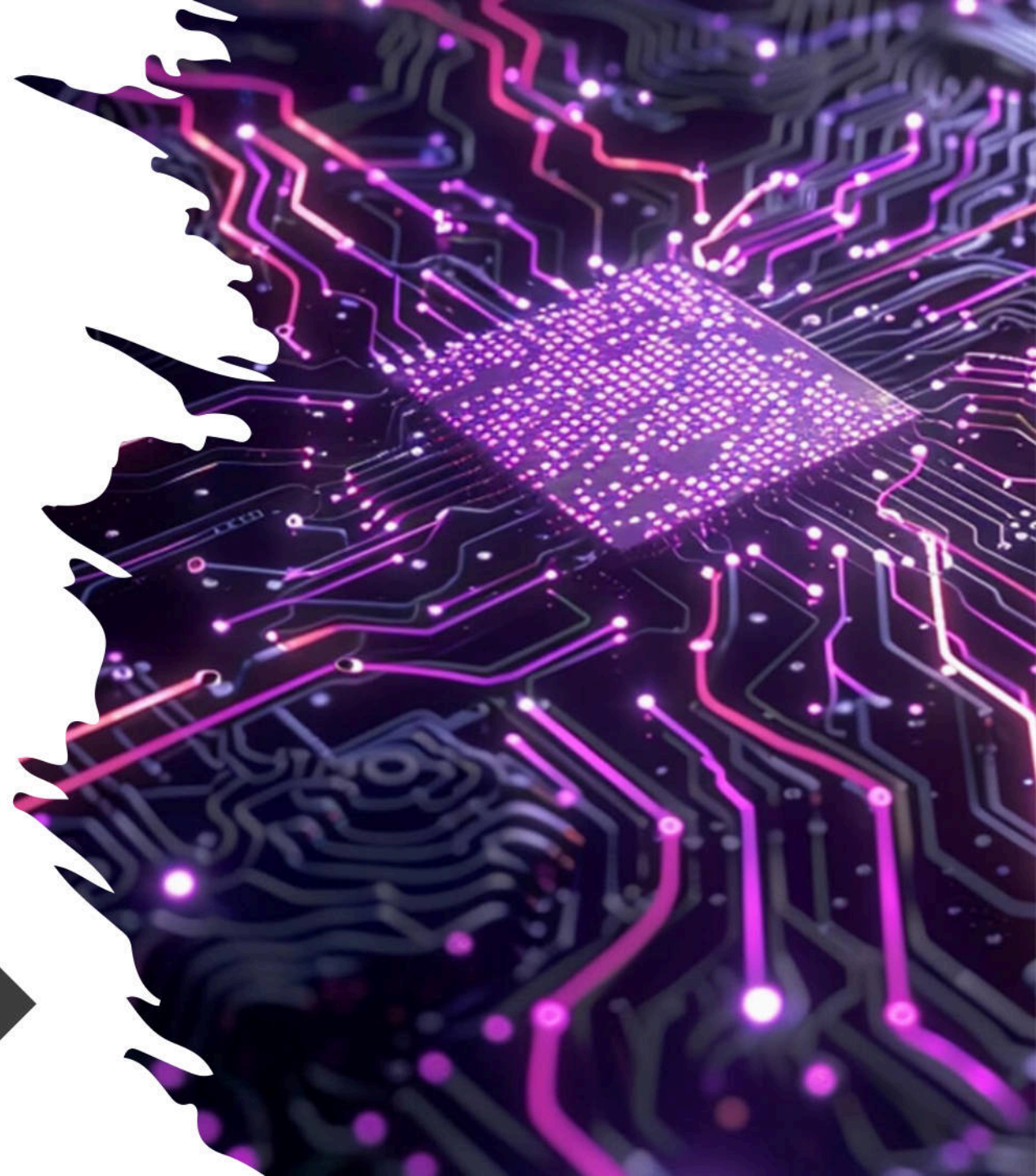
Demos & Defences

Finding the
Side Channel

Token-Length
Extraction

Exploitation
with AI

Demos &
Defences



Putting it All together

- Training an LLM from scratch
 - Takes months
 - Costs millions \$\$\$
- We followed a common strategy
 - Finetuning

Our Resources

Pretrained Model



T5-flan

Dataset for Finetuning

UltraChat

Libraries (Python)



Hugging Face

Our Resources

- 2 days of compute time
- ~200\$ on Azure for example



Copilot with GPT-4

https://www.bing.com/chat?form=NTPCHB

Microsoft Bing SEARCH COPILOT Sign in

 **Copilot**

Your everyday AI companion

with  Bing

 **Create**
Create a slogan for a new social media platform that specializes in sarcasm.

 **Organize**
Organize the last ten years of worldwide volcanic activity into a table.

 **Shop**
What are some popular children's clothing brands?

 **Write**
Write step-by-step instructions to make pizza crust and give me the ingredient list.

 **Travel**
Where do people travel for culinary experiences?

 **Code**
Write a Python script to perform Binary search.

 **Compare**
I'm thinking of making a career change. Can you help me go through pros and cons?

Bing is powered by AI, so surprises and mistakes are possible. Please share feedback so we can improve! [Terms](#) | [Privacy](#) | [FAQs](#)

Choose a conversation style

More Creative

More Balanced

More Precise

[Sign in](#) to ask more questions and have longer conversations

Ask me anything...

0/4000



The Wireshark Network Analyzer

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

Welcome to Wireshark

Open

C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\public_schools\public_schools.pcapng (5215 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\disabilities\disabilities.pcapng (3667 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\covid\covid.pcapng (6451 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\mlai\mlai.pcapng (4287 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\ai\ai.pcapng (4335 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\renewable_energy\energy.pcapng (4557 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\agricultural\agriculture.pcapng (4268 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\BingAI\public_schools.pcapng (not found)
C:\Users\Daniel\Desktop\ChatLogger\Data\KaliCron\hour_half\output_20240201_180001\conversation_7.pcapng (10304 KB)
C:\Users\Daniel\Desktop\ChatLogger\Data\AutoSniff_Lab\LAN1\conversation_1.pcap (10388 KB)

Capture

...using this filter: All interfaces shown

Wi-Fi
Ethernet 2
Ethernet 3
Adapter for loopback traffic capture
Local Area Connection* 10
Local Area Connection* 9
Local Area Connection* 8
Bluetooth Network Connection
Local Area Connection* 2
Local Area Connection* 1
Ethernet

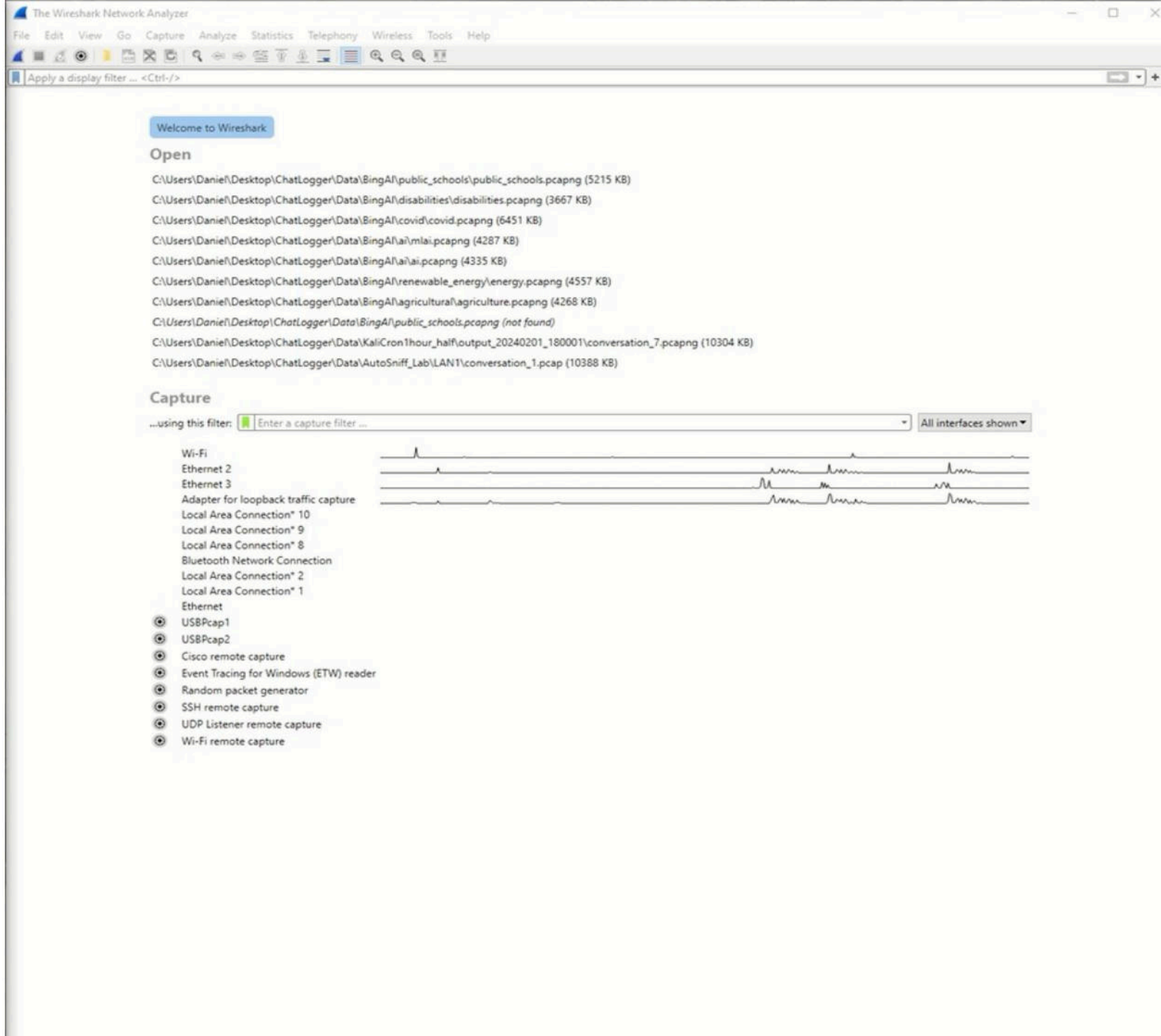
☒ USBPcap1
☒ USBPcap2
☒ Cisco remote capture
☒ Event Tracing for Windows (ETW) reader
☒ Random packet generator
☒ SSH remote capture
☒ UDP Listener remote capture
☒ Wi-Fi remote capture

Learn

User's Guide · Wiki · Questions and Answers · Mailing Lists · Sharkfest · Wireshark Discord · Donate

You are running Wireshark 4.2.0 (v4.2.0-0-g54eedf63953). You receive automatic updates.

Ready to load or capture No Packets Profile: Default



Copilot with GPT-4

https://www.bing.com/chat?form=NTPCHB

Microsoft Bing

SEARCH

COPILOT

Sign in



Copilot

Your everyday AI companion

with  Bing

 **Create**
Create a slogan for a new social media platform that specializes in sarcasm

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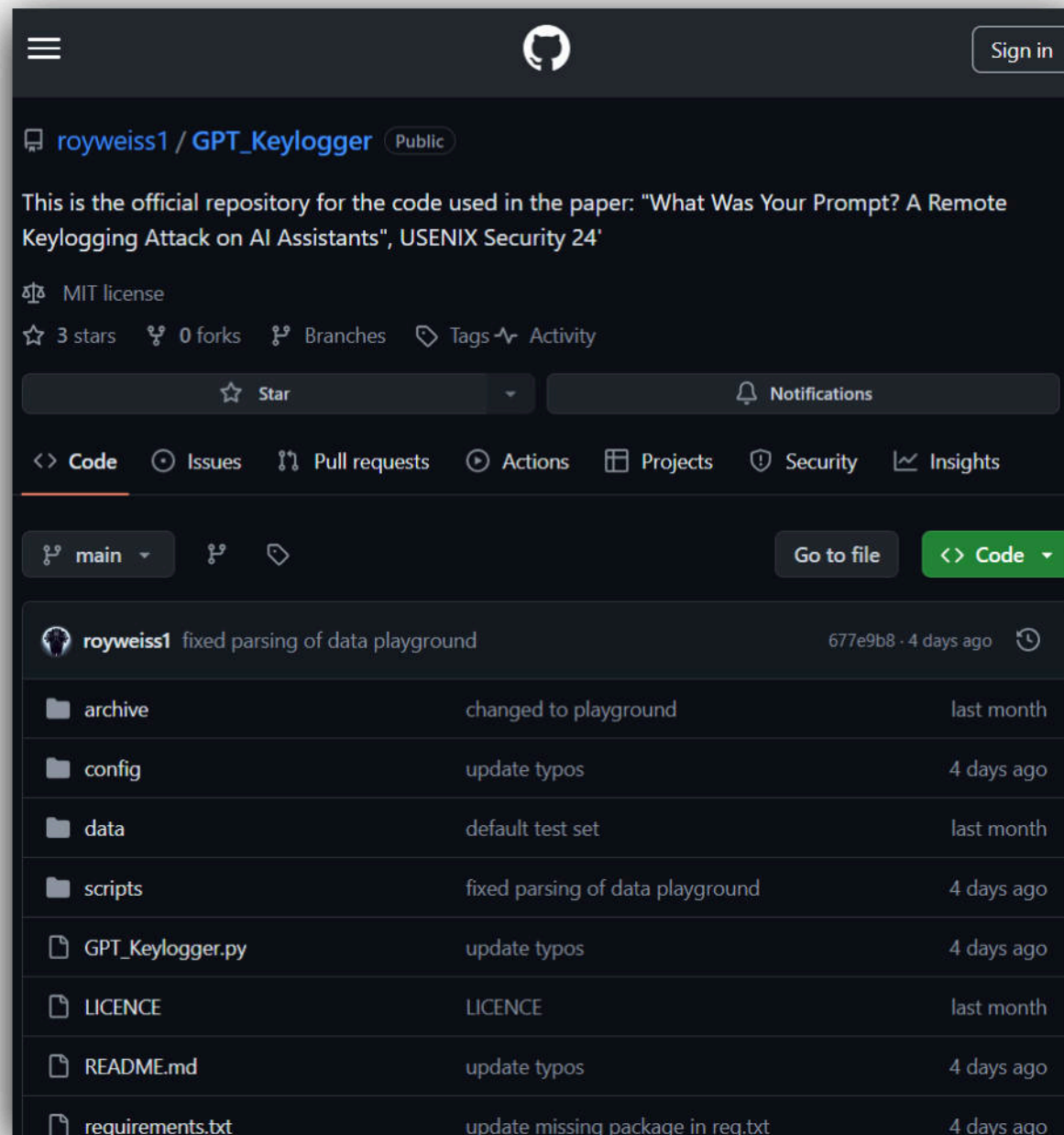
The Wireshark N

File Edit View

Apply a display f

Try it yourselves!

- We have developed a full Click & Play tool:
- https://github.com/royweiss1/GPT_Keylogger



GPT-Keylogger



Features:

- **Attack:** token-length-sequence > text
- **Trainer:** Finetuning a model to other datasets

Assets

- Model trained on ChatGPT4 responses (Feb 2024)

GPT-Keylogger: *Attack*



- Can receive as input:
 - Packet Lengths – will calculate the Deltas
 - Token Lengths
 - Original Text – will tokenize and predict

```
python GPT_Keylogger.py --playground
```

GPT-Keylogger: *Trainer*



- Every LLM is different
- Train your own model on your target LLM!
- Default Configuration file at the GitHub repository.

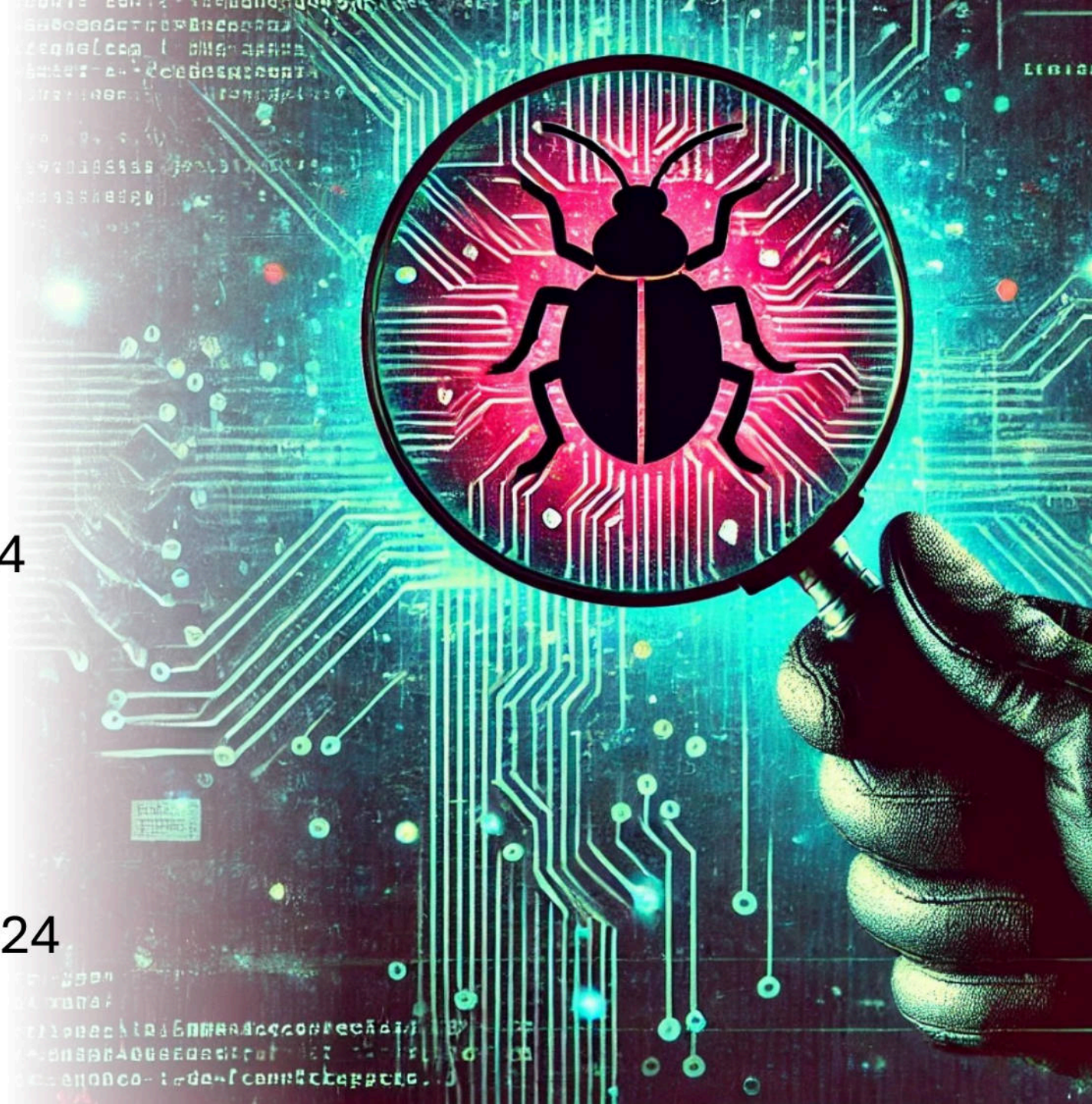
```
python GPT_Keylogger.py --train-first-sentences  
<configuration file>
```

```
python GPT_Keylogger.py --train-middle-sentences  
<configuration file>
```


Disclosure ● Feb 8th 2024

Patch ● March 1st 2024

Public Release ● March 14th 2024
Ars Technica ↓





Defences

1. Random Padding
2. Padding to the nearest
3. More Buffering

Defences

1. Random Padding
2. Padding to the nearest
3. More Buffering



Padding to the closest - ChatGPT



```
data: {"message": {"id": "0a5e903b-a872-4231-8817-98b889c93b93", "author": {"role": "assistant", "name": null, "metadata": {}},
"create_time": 1720961039.676914, "update_time": null, "content": {"content_type": "text", "parts":
["In the realm where shadows dance with light,\nBeneath the silver moon's soft, gentle might,\nA"]}
, "status": "in_progress", "end_turn": null, "weight": 1.0, "metadata": {"citations": [], "gizmo_id": null, "message_type": "next",
"model_slug": "gpt-4o", "default_model_slug": "gpt-4o",
"pad": "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA",
"parent_id": "0e1b35b6-ba4a-4e1e-a963-86ed4901c118", "model_switcher_deny": [], "recipient": "all", "channel": null},
"conversation_id": "9181d8ac-a77d-4905-aaf1-1bb04cf1d4a8", "error": null}
```

Padding to the closest - ChatGPT



1685	15.987522	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1688	16.009442	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1691	16.031254	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1694	16.039914	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1697	16.072241	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1700	16.076127	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1703	16.146251	172.64.155.141	10.100.102.8	WebSocket	72	WebSocket	Text	[FIN]
1708	16.165411	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1714	16.179877	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1717	16.199583	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1720	16.215137	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1723	16.230746	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1726	16.252399	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1729	16.272455	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1732	16.287093	172.64.155.141	10.100.102.8	WebSocket	116	WebSocket	Text	[FIN]
1735	16.306392	172.64.155.141	10.100.102.8	WebSocket	100	WebSocket	Text	[FIN]
1742	16.321668	172.64.155.141	10.100.102.8	WebSocket	160	WebSocket	Text	[FIN]
1746	16.345858	172.64.155.141	10.100.102.8	WebSocket	160	WebSocket	Text	[FIN]
1749	16.360370	172.64.155.141	10.100.102.8	WebSocket	160	WebSocket	Text	[FIN]
1752	16.377564	172.64.155.141	10.100.102.8	WebSocket	160	WebSocket	Text	[FIN]
1755	16.394136	172.64.155.141	10.100.102.8	WebSocket	160	WebSocket	Text	[FIN]



Random Padding - CloudFlare

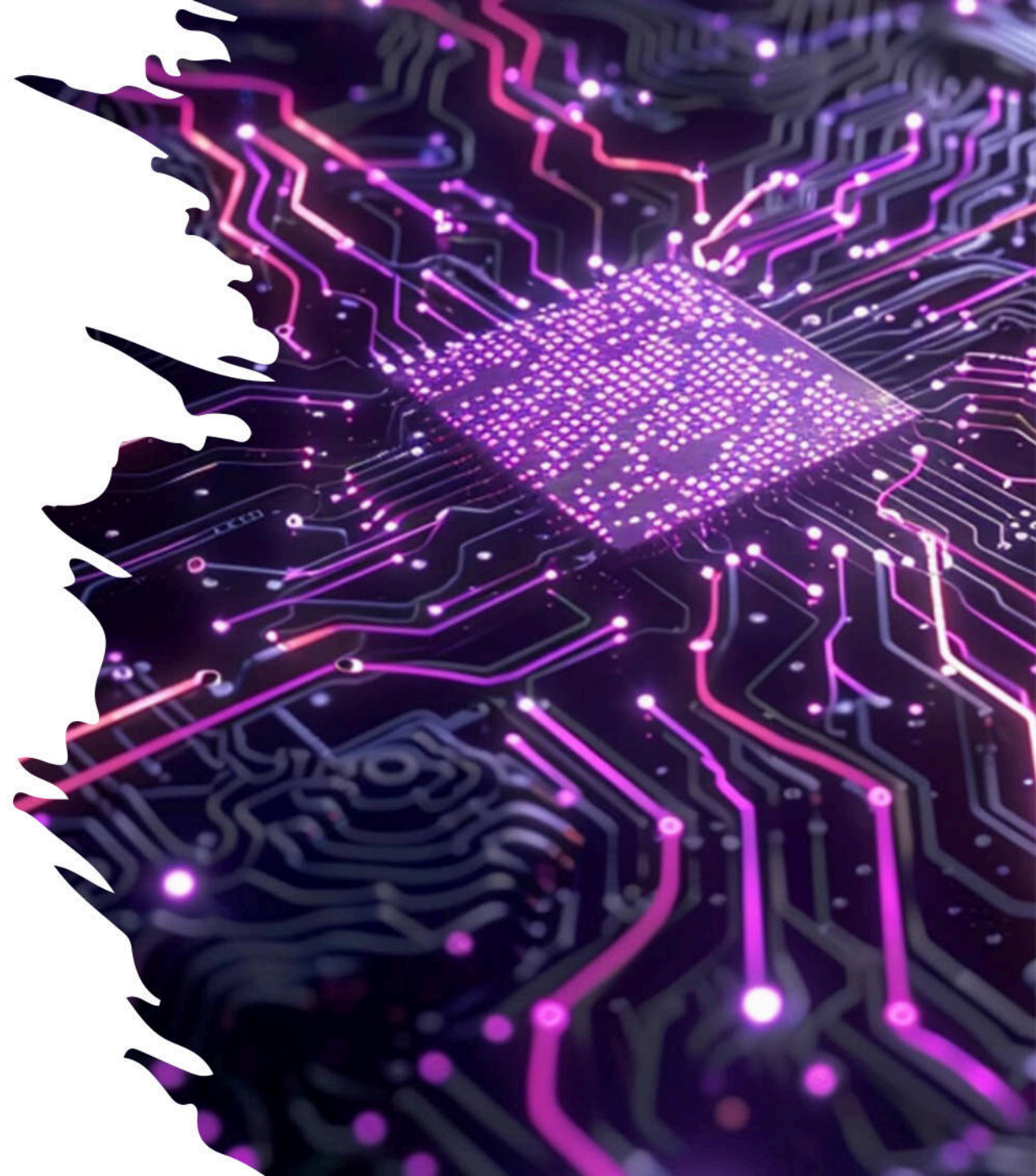
```
data: {"response": " Machine", "p": "ab"}  
data: {"response": " learning", "p": "abcdefghijklmnopqrstuvwxyz0123456789abcdefghijklmnopqrstuvwxyz"}  
data: {"response": " and", "p": "abcdef"}  
data: {"response": " artificial", "p": "abcdefghij"}  
data: {"response": " intelligence", "p": "abcdefghijklmnopqrstuvwxyz"}  
data: {"response": " (", "p": "abcdefghijklmnopqrstuvwxyz0123456789abcdefghijklmnopqrstuvwxyz"}  
data: {"response": " AI", "p": "abcdefghijklmnopqrstuvw"}  
data: {"response": " )", "p": "a"}  
data: {"response": " have", "p": "abcdefghijklmnopqrstuvwxyz0123456789abcdefg"}  
data: {"response": " the", "p": "abcdefghijklmnopqrstuvw"}  
data: {"response": " potential", "p": "abcdefghi"}  
data: {"response": " to", "p": "abcdefghijklmnopqrstuvwxyz01234567"}  
data: {"response": " revolution", "p": "abcdefghijklmnopqrstuvwxyz012"}  
data: {"response": " ize", "p": "abcdefghijklmnopqrstuvwxyz0123456789a"}  
data: {"response": " the", "p": "abcdefghijklmnopqrs"}  
data: {"response": " health", "p": "abcdefghijklmnopqrstuvwxyz012345"}  
data: {"response": " care", "p": "abcdefghijklmnopqrstuvwxyz0123456789"}  
data: {"response": " industry", "p": "abcdefghijklmnopqrstuvwxyz"}
```

Random Padding - CloudFlare



88	5.058415000	104.19.192.175	10.100.102.12	TLSv1.3	122	[TLS segment of a reassembled PDU]
91	5.113880000	104.19.192.175	10.100.102.12	TLSv1.3	167	[TLS segment of a reassembled PDU]
93	5.173536000	104.19.192.175	10.100.102.12	TLSv1.3	122	[TLS segment of a reassembled PDU]
95	5.223983000	104.19.192.175	10.100.102.12	TLSv1.3	133	[TLS segment of a reassembled PDU]
98	5.277632000	104.19.192.175	10.100.102.12	TLSv1.3	150	[TLS segment of a reassembled PDU]
101	5.333802000	104.19.192.175	10.100.102.12	TLSv1.3	158	[TLS segment of a reassembled PDU]
104	5.390742000	104.19.192.175	10.100.102.12	TLSv1.3	137	[TLS segment of a reassembled PDU]
106	5.445842000	104.19.192.175	10.100.102.12	TLSv1.3	114	[TLS segment of a reassembled PDU]
110	5.498872000	104.19.192.175	10.100.102.12	TLSv1.3	160	[TLS segment of a reassembled PDU]
112	5.554440000	104.19.192.175	10.100.102.12	TLSv1.3	140	[TLS segment of a reassembled PDU]
114	5.608645000	104.19.192.175	10.100.102.12	TLSv1.3	131	[TLS segment of a reassembled PDU]
116	5.670327000	104.19.192.175	10.100.102.12	TLSv1.3	149	[TLS segment of a reassembled PDU]
119	5.718106000	104.19.192.175	10.100.102.12	TLSv1.3	152	[TLS segment of a reassembled PDU]
121	5.772857000	104.19.192.175	10.100.102.12	TLSv1.3	152	[TLS segment of a reassembled PDU]
123	5.827343000	104.19.192.175	10.100.102.12	TLSv1.3	135	[TLS segment of a reassembled PDU]
125	5.881446000	104.19.192.175	10.100.102.12	TLSv1.3	151	[TLS segment of a reassembled PDU]
128	5.940902000	104.19.192.175	10.100.102.12	TLSv1.3	152	[TLS segment of a reassembled PDU]
129	5.991357000	104.19.192.175	10.100.102.12	TLSv1.3	110	[TLS segment of a reassembled PDU]
130	5.991357000	104.19.192.175	10.100.102.12	TLSv1.3	119	[TLS segment of a reassembled PDU]
135	6.047785000	104.19.192.175	10.100.102.12	TLSv1.3	157	[TLS segment of a reassembled PDU]
136	6.103934000	104.19.192.175	10.100.102.12	TLSv1.3	167	[TLS segment of a reassembled PDU]
140	6.156732000	104.19.192.175	10.100.102.12	TLSv1.3	124	[TLS segment of a reassembled PDU]
142	6.214439000	104.19.192.175	10.100.102.12	TLSv1.3	121	[TLS segment of a reassembled PDU]

Conclusion



Summary

- New side channel: Token-length
- First use of GenAI to read a side channel
- Many vendors affected
 - Some patched... 🙄
 - What about new services?

Key Takeaways

1. Make sure to use encryption schemes correctly
 - AI services are no exception!
2. AI is your friend
 - What seemed impossible is now possible
 - What other side channels can be broken with AI?

Thank You!

Yisroel Mirsky, Roy Weiss,
Daniel Ayzenshteyn, Guy Amit

*Offensive AI Research Lab
Ben-Gurion University*

<https://offensive-ai-lab.github.io/>

